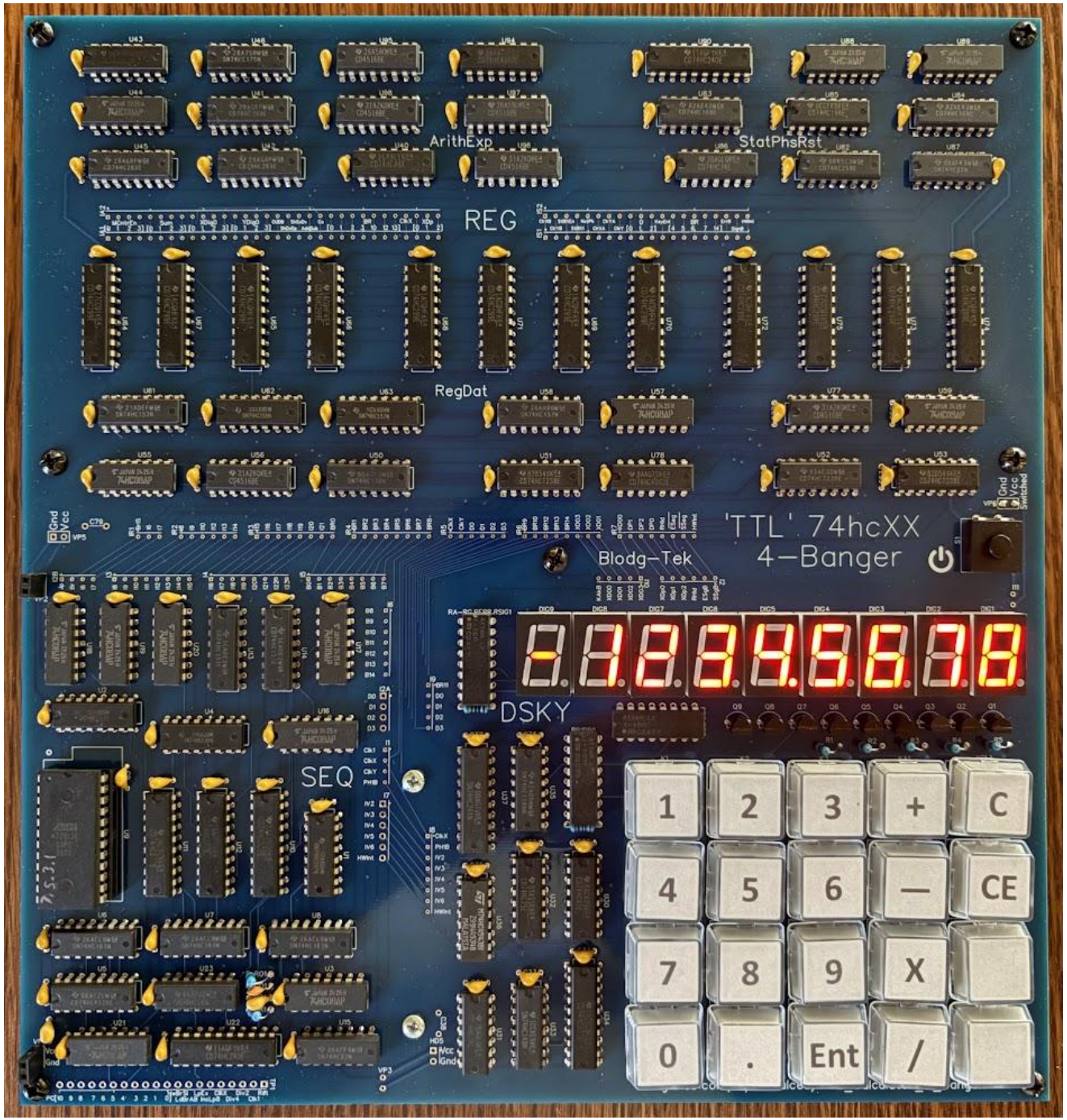
A “TTL” 4-Function Calculator Overview

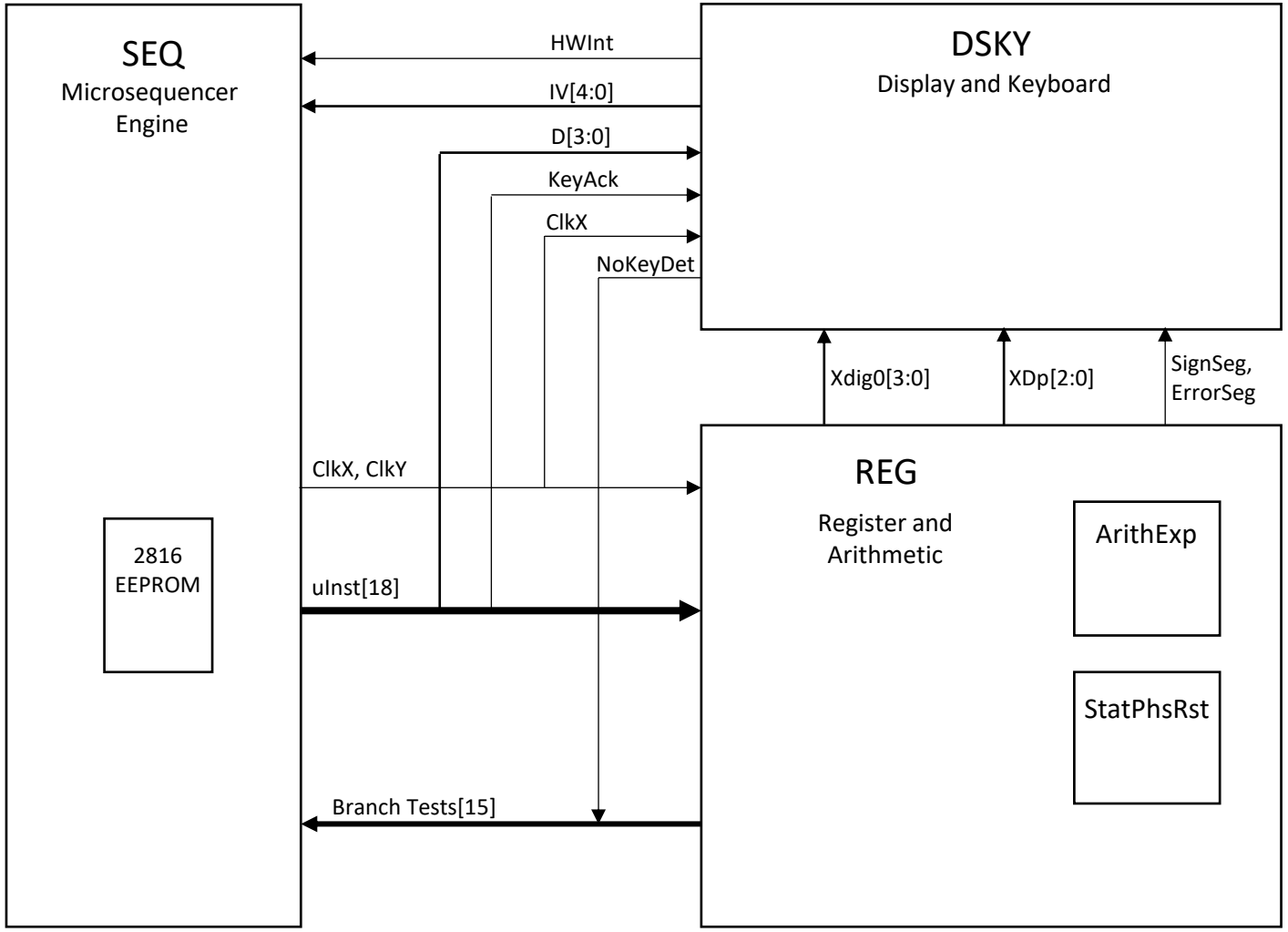
Specs

- 9x10” PCB, 4 layers
- All 74hcXX logic implementation (plus 3 4XXX) chips, 76 total, and one 2816 EEPROM
- Purpose built architecture (not a CPU architecture)
- RPN style operation (HP Enter key)
- Add, subtract, multiply, and divide, clear entry, clear all
- 8 numeric digits plus an Error and sign digit
  - Max num: +99,999,999 to -99,999,999
  - Min Num: +0.0000001 to -0.0000001
- Floating decimal point
- Signals error on overflow for all operations, divide by 0, and keyboard input overflow
- Leading zero blanking during operand entry
- Running total operation (last result = Operand1 + Enter)
- No rounding logic or guard digits

https://github.com/Sphaeralcea/TTL_Calculator_4_Banger



Top Level


SEQ
Microsequencer
Engine
2816
EEPROM
HWInt
IV[4:0]
D[3:0]
KeyAck
ClkX
NoKeyDet
ClkX, ClkY
uInst[18]
Branch Tests[15]
DSKY
Display and Keyboard
Xdig0[3:0]
XDp[2:0]
SignSeg,
ErrorSeg
REG
Register and
Arithmetic
ArithExp
StatPhsRst

Overview

3 Major logic blocks

- SEQ: Micro-sequencer engine with 2Kx8b uROM
  - 24-bit micro-instruction word
  - Unconditional branch, 16 conditional branches, 1 NMI-type interrupt
  - Single level loop counter (no nesting)
  - Clock and reset generation
- DSKY: Keyboard and Display unit
  - 9-digit LED display sequencing,
  - 18(20)-key keyboard encoding (2 keys not used)
- REG: Register-Arithmetic unit
  - 3, 8-digit registers (4-bits X 8)
  - 1 digit BCD adder/subtractor and exponent calculator (counters)
  - 7 status and control register bits, operation progress tracking state machine

- Architectural deficits (intentional)
  - No rounding logic, guard digits
- Architectural bugs (unintentional)
  - Accuracy loss dividing 2 8-digit integers

Top Level Process View

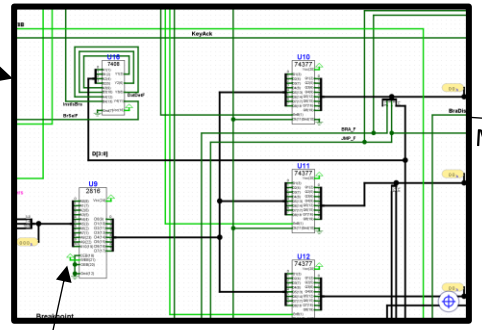
Design Entry

Schematic Entry

Analog Parts

uCode Entry

Logisim Evolution
RTL Simulation Schematic

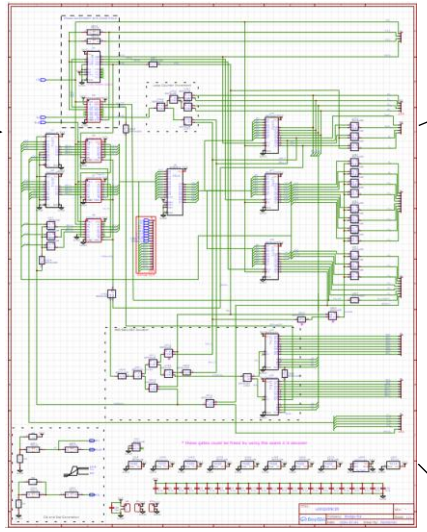


Design Feedback

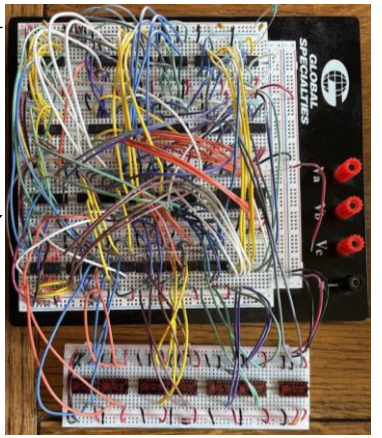
Manual Sch Translation ☹

Manual Formal Verification ☹

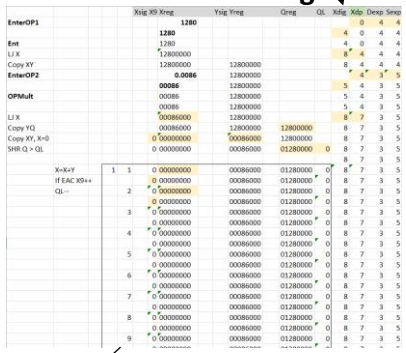
EasyEDA std/JLCPCB



BreadBoard



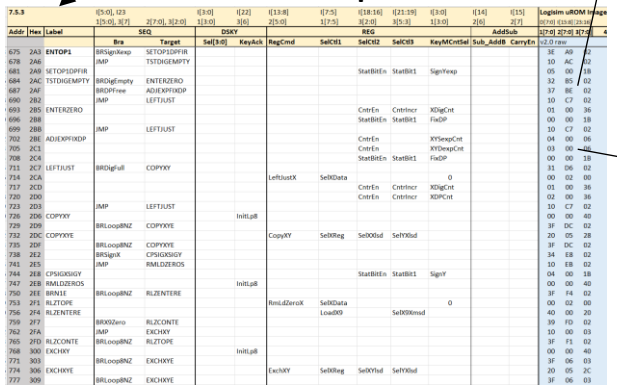
uCode Modeling



LTSpice

Disp
Clk
Rst

uCode Assembler Spreadsheet

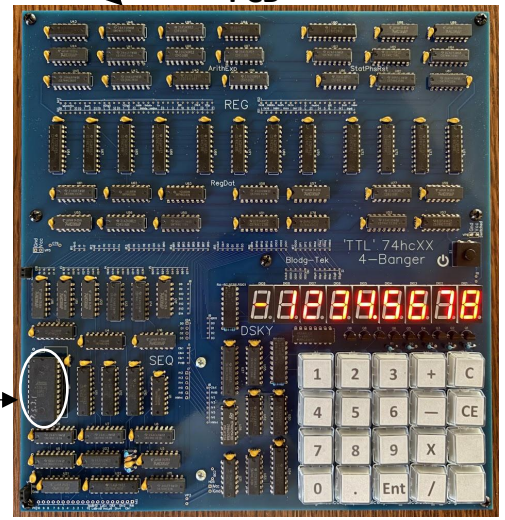


Txt File

Arduino Sketch

2816 Programmer

PCB



- Component placement is manual
- Autoroute used to route the modules
- Manual route to stich together modules

SEQ

Microsequencer

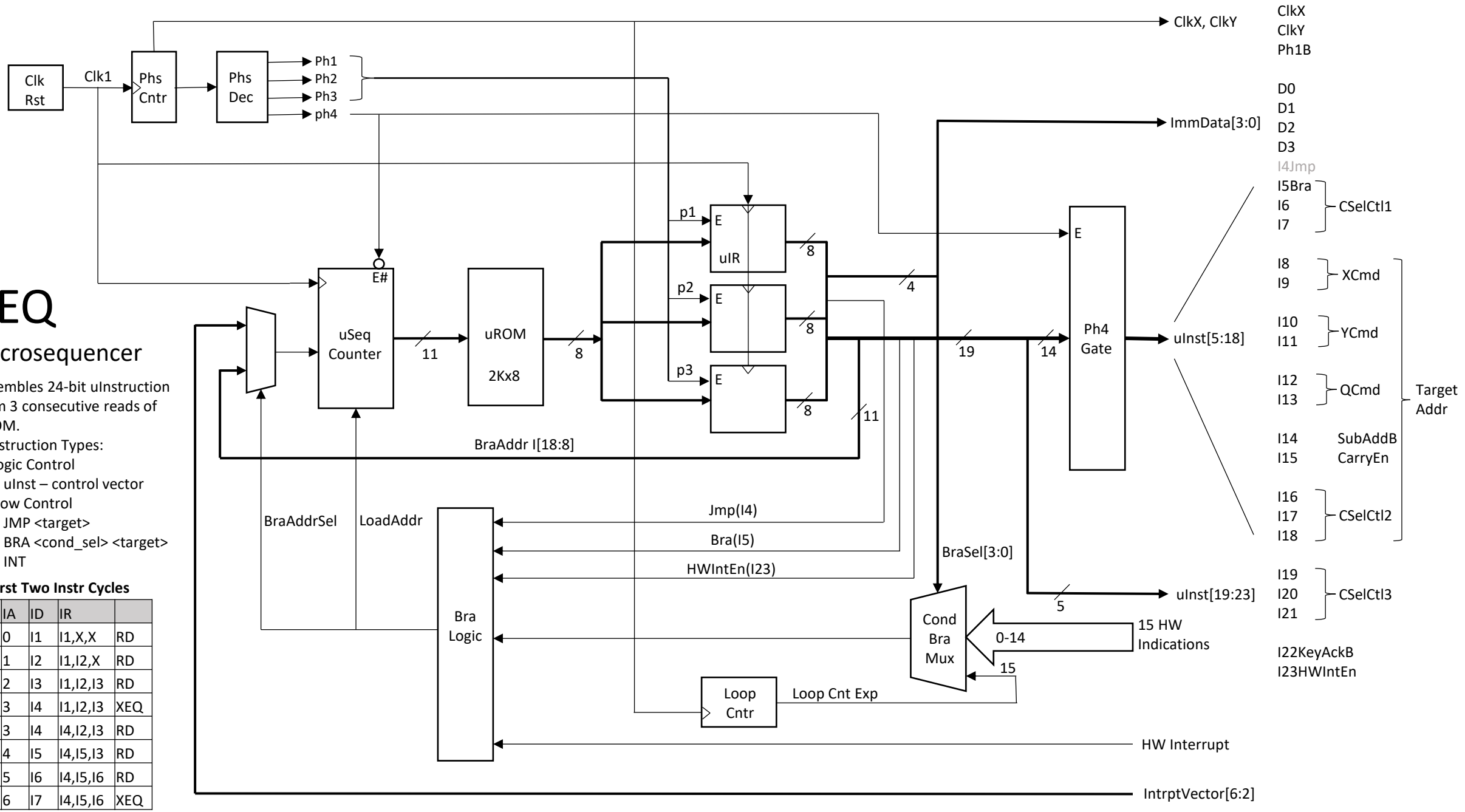
Assembles 24-bit uInstruction from 3 consecutive reads of uROM.

4 Instruction Types:

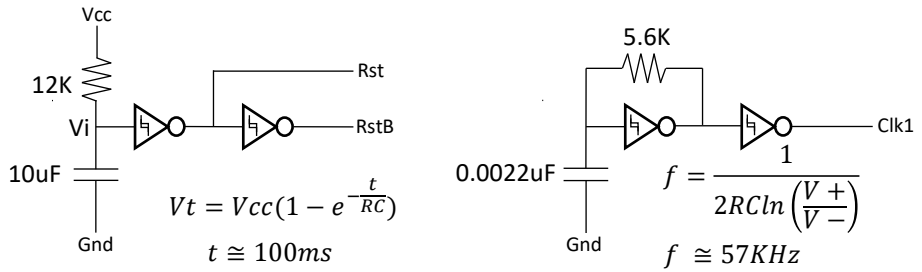
- Logic Control
  - uInst – control vector
- Flow Control
  - JMP <target>
  - BRA <cond_sel> <target>
  - INT

First Two Instr Cycles

Phs	IA	ID	IR	
P1	0	I1	I1,X,X	RD
P2	1	I2	I1,I2,X	RD
P3	2	I3	I1,I2,I3	RD
P4	3	I4	I1,I2,I3	XEQ
P1	3	I4	I4,I2,I3	RD
P2	4	I5	I4,I5,I3	RD
P3	5	I6	I4,I5,I6	RD
P4	6	I7	I4,I5,I6	XEQ



Power-On Reset Startup

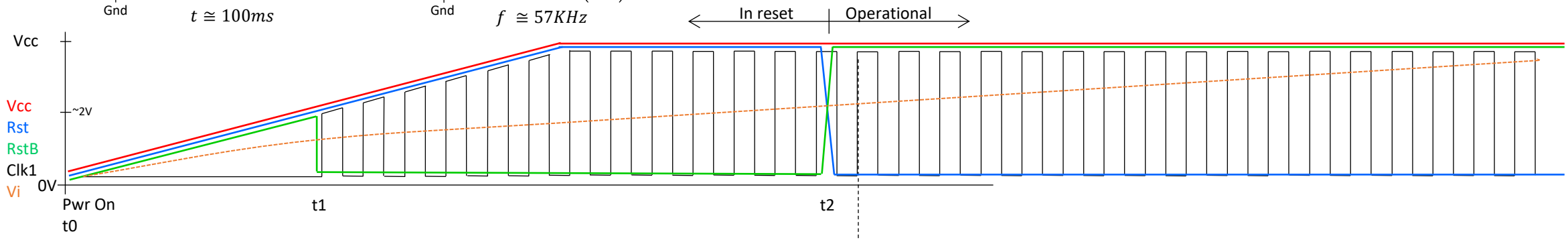


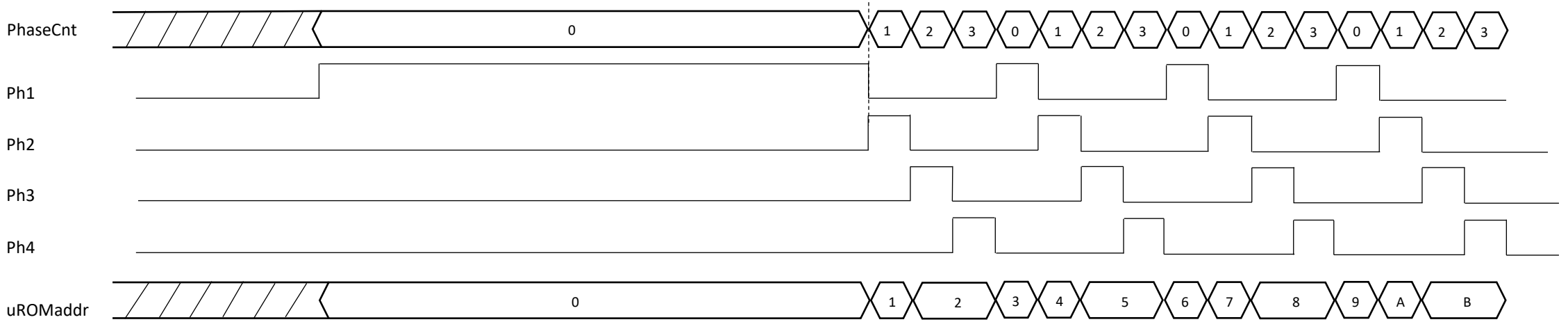
$$Vt = Vcc(1 - e^{-\frac{t}{RC}})$$

$$t \cong 100ms$$

$$f = \frac{1}{2RCln\left(\frac{V+}{V-}\right)}$$

$$f \cong 57KHz$$





Sequencer (SEQ)

- Includes Clock, Reset and Timing generation
  - Schmitt trigger reset holds sequencer in reset until power rises to ~75% VCC
  - Schmitt trigger oscillator generates 1X clock
  - 1X clock drives SEQ logic only. Does not clock other units
  - 1X clock feeds the uInst PC, the IR, and the phase counter which also generates the div4 clocks, ClkX and ClkY
  - ClkX and ClkY clock the other units
- Sequencer operates on a 4-phase instruction cycle to construct and present a 24-bit micro-instruction, 23 of which are available to control external hardware
  - First 3 phases read three consecutive bytes from uROM and stores them in 3 8-bit registers
  - During first 3 phases, external hardware sees 0's (inactive) on all signals
  - Fourth phase ungates the uInstruction and presents it to external hardware
  - D0 – D3: Immediate data and select field
  - I4: Instruction type. This bit is reserved for the sequencer's exclusive use
    - I4=0: Instruction is a hardware control instruction
    - I4=1: Instruction is a flow control instruction (JMP or BRA)

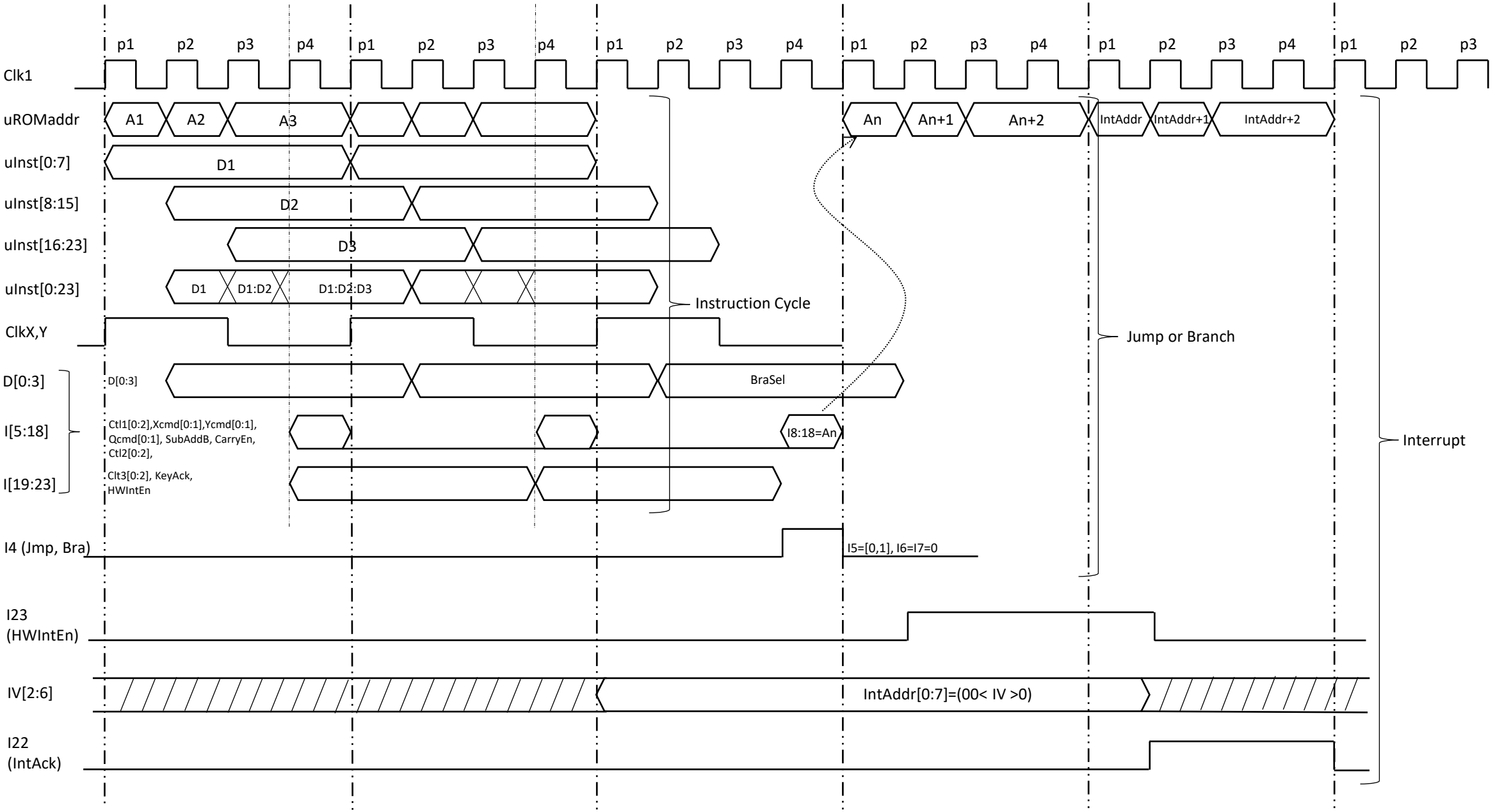
Sequencer (SEQ)

- I5: HW control and flow control qualifier
  - If I4=0, then I5 is just another hardware control bit
  - If I4=1, then I5 is flow control qualifier
    - I5=0, unconditional (JMP)
    - I5=1, conditional (BRA)
      - One of 16 conditions is tested, selected by D[3:0]. If condition is true then branch is taken
      - D[3:0]=0xF is reserved. Dedicated internally to implement a simple for-loop capability
- I6 – I21: uINSTR bits
- I22: Loop Init: Used internal to SEQ to reset a counter that implements a simple for-loop capability
  - I22 is also used as external to SEQ to register the state of keyboard (KeyAckB)
- I23: Interrupt enable. Used internal to SEQ to enable a hardware-initiated branch. Taken if external hardware has asserted the HWInt signal
  - External hardware provides the target address through a 5-bit vector, IV[6:2]
  - The target address is formed by appending IV[6:2] with b00
    - Hardware jump targets are confined to uROM addresses at 4-byte boundaries between 0x10 to 0x7C
  - I23 is available external to SEQ but not used

Sequencer (SEQ)

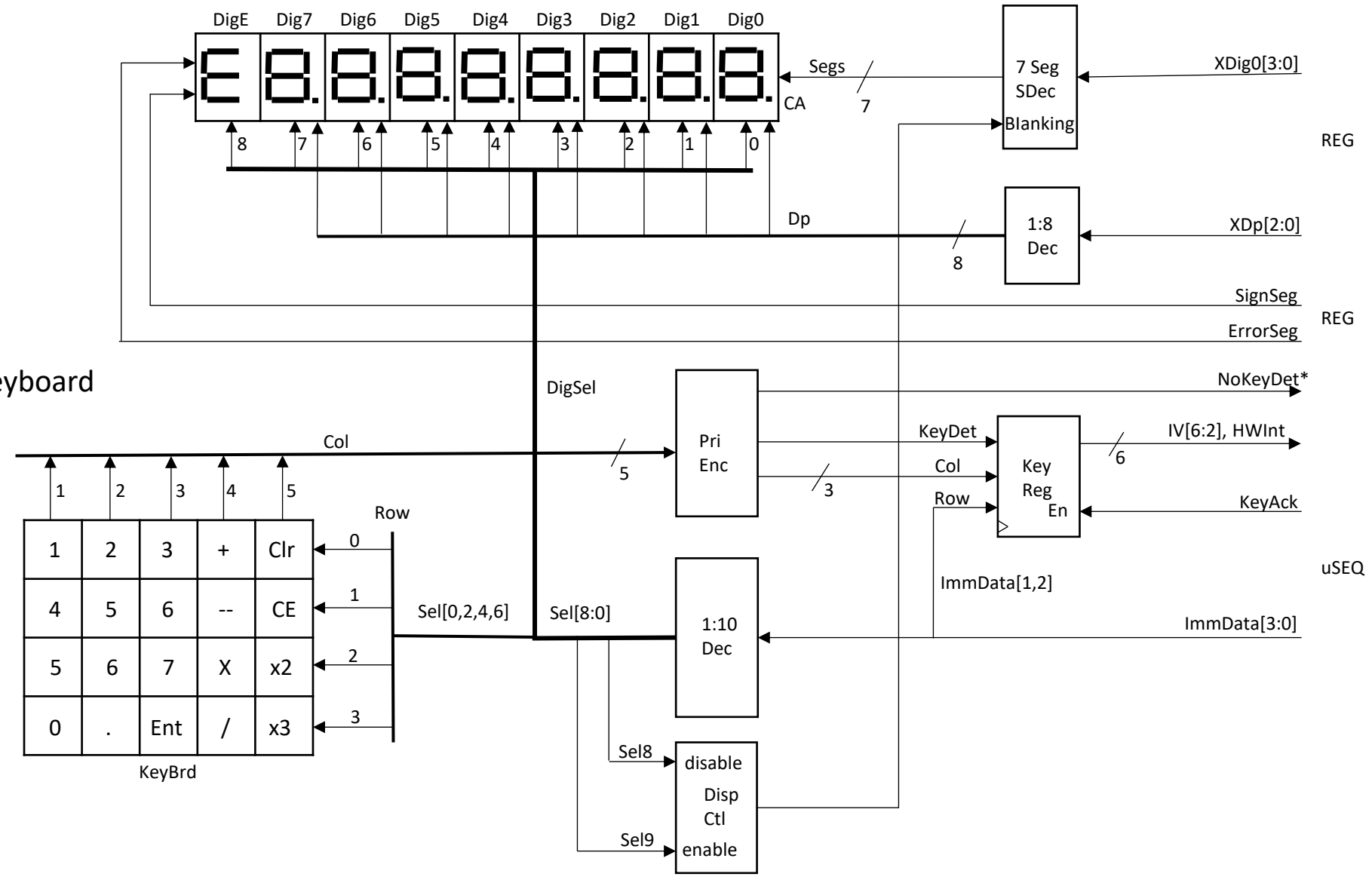
- A primitive looping structure provides the ability to create loops up to 9 iterations (0-8)
  - Loop counter reset by asserting I22 (InitLP8)
  - Loop count tested and incremented by a conditional jump with D[3:0]=0xF (BRLoop8NZ)
  - Loop count expires when loop count reaches 8, and loop will fall through on the next loop count test (BRLoop8NZ)
- Note: There is no subroutine call and return capability provided.
  - Since all keyboard service routines return to the keyboard-display loop, an unconditional JMP to this loop serves as an adequate return from keyboard service routines.

Sequencer Instruction Timing



DSKY

Display and Keyboard



* Branch Condition

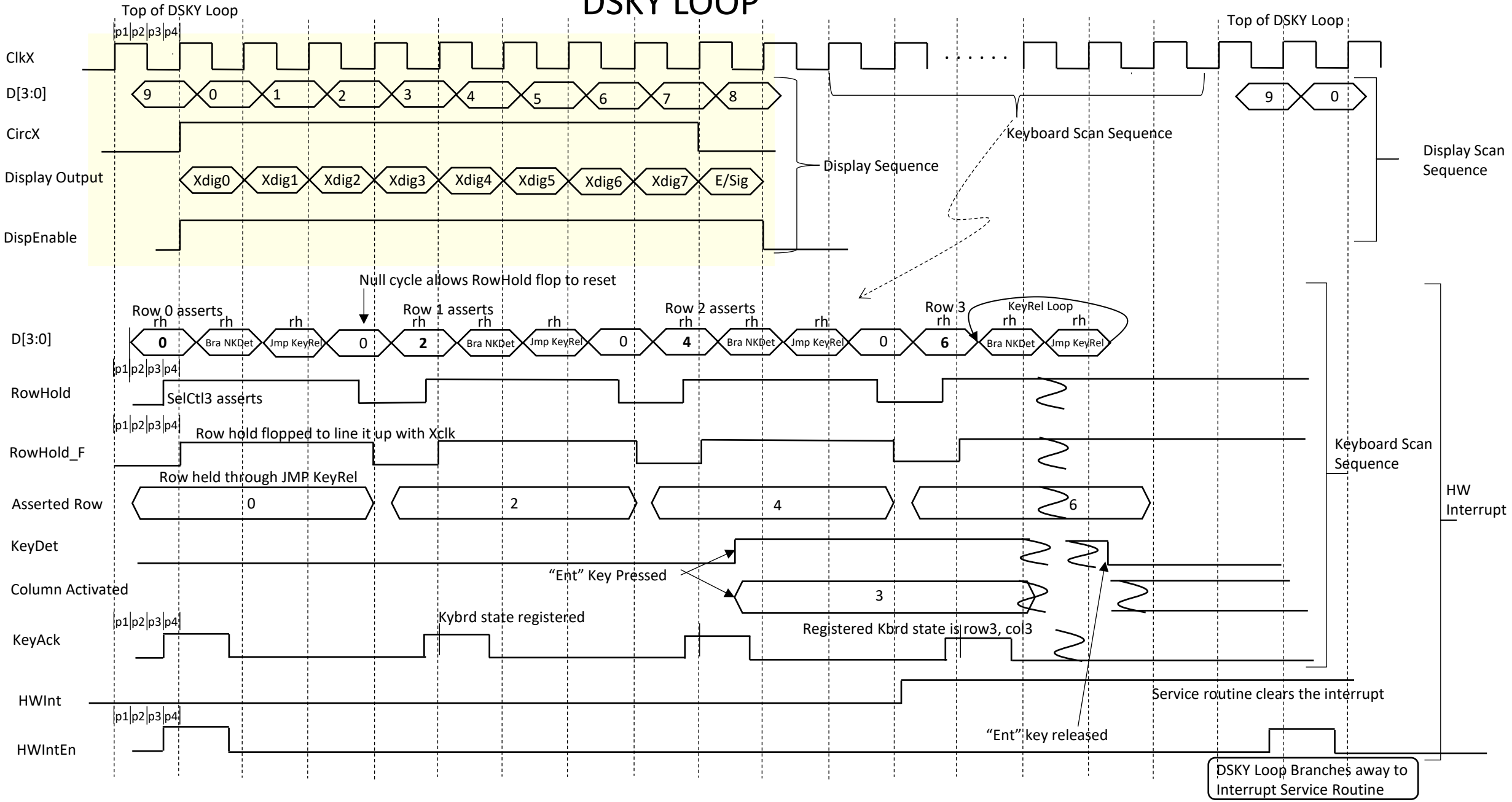
Scan Sequence

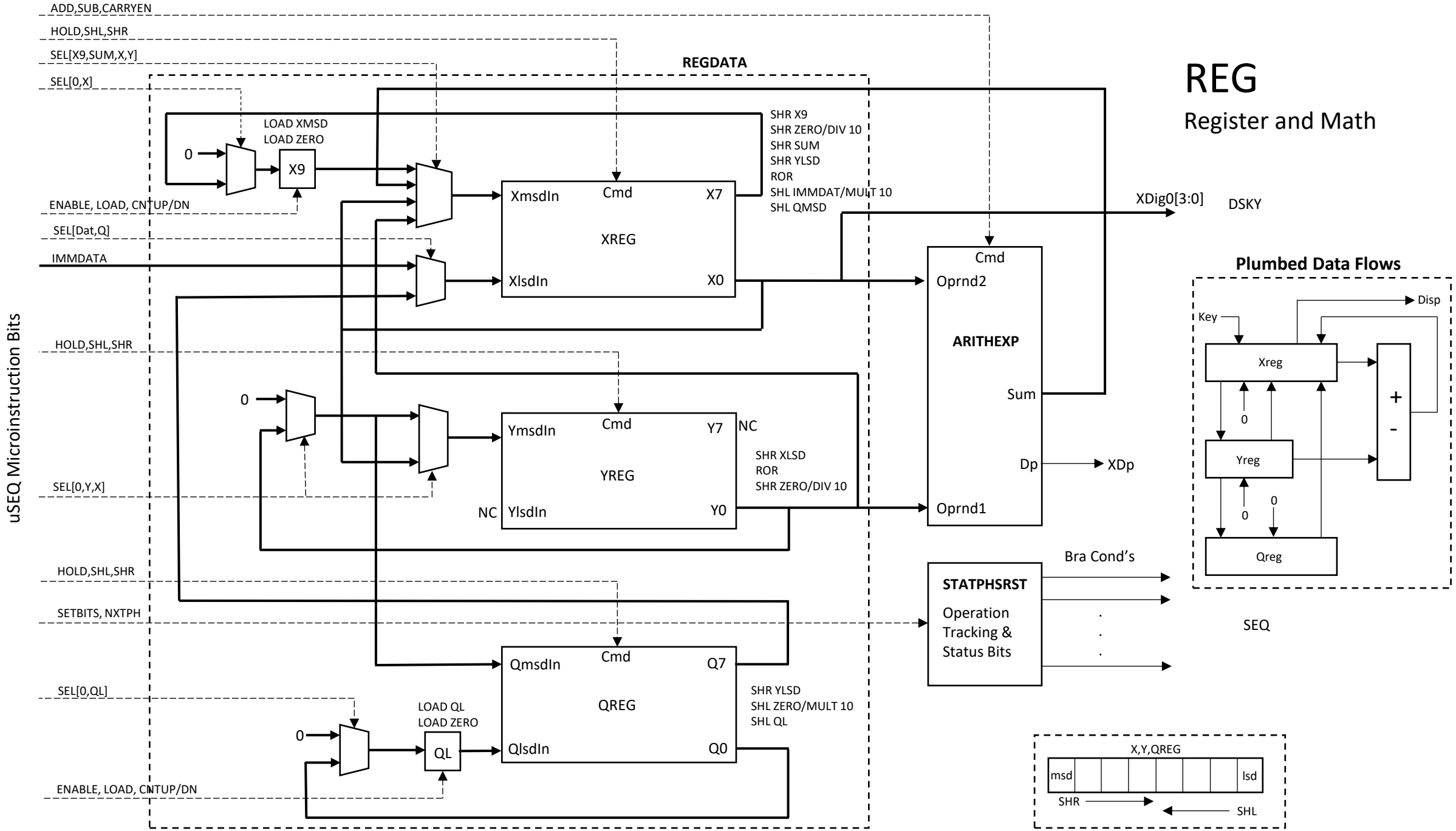
9 – Disp En, IntEn
0 – Dig0
1 – Dig1
2 – Dig2
3 – Dig3
4 – Dig4
5 – Dig5
6 – Dig6
7 – Dig7
8 – Dig8; Disp Dis
0 – KeyRow1, KeyAck
2 – KeyRow2, KeyAck
4 – KeyRow3, KeyAck
6 – KeyRow4, KeyAck

Display and Keyboard (DSKY)

- DSKY is a passive module with no autonomy. Completely controlled by SEQ
- Display and keyboard scan is performed via D[3:0] from SEQ
  - The display is enabled by D[3:0]=b1001, and it is disabled by b1000
  - Dig0 is activated by D[3:0]=b0000, Dig7 by b0111, the Sign/Error digit by b1000
  - Row 0 of the keyboard matrix is activated by b0000, row 1 by b0010, row 2 by 0100, and row 3 by b0110. The display is disabled during row scan
- REG provides BCD data for each digit to display, 3-bit decimal point position, and the sign and error segments
- When a key is pressed in an activated row, a key detect signal is raised
- If microcode running on SEQ tests the detect signal to be true, then it may assert the KeyAck (I22) signal back to DSKY which registers the row and column where the depressed key resides and asserts the HWInt signal
- On any microcode instruction that asserts HWIntEn (I23) while HWInt is asserted, SEQ flow will jump to the uROM address formed by the concatenation of col[2:0], row[1:0], and b’00

DSKY LOOP


Top of DSKY Loop
p1 p2 p3 p4
ClkX
D[3:0]
9 0 1 2 3 4 5 6 7 8
CircX
Display Output
Xdig0 Xdig1 Xdig2 Xdig3 Xdig4 Xdig5 Xdig6 Xdig7 E/Sig
DispEnable
Display Sequence
Top of DSKY Loop
9 0
Keyboard Scan Sequence
Display Scan Sequence
Null cycle allows RowHold flop to reset
Row 0 asserts
Row 1 asserts
Row 2 asserts
Row 3
KeyRel Loop
rh
0 Bra NKDet Jmp KeyRel 0 2 Bra NKDet Jmp KeyRel 0 4 Bra NKDet Jmp KeyRel 0 6 Bra NKDet Jmp KeyRel
RowHold
SelCtl3 asserts
RowHold_F
Row hold flopped to line it up with Xclk
Row held through JMP KeyRel
Asserted Row
0 2 4 6
KeyDet
"Ent" Key Pressed
Column Activated
3
Kybrd state registered
KeyAck
Registered Kbrd state is row3, col3
Keyboard Scan Sequence
HW Interrupt
HWInt
Service routine clears the interrupt
"Ent" key released
HWIntEn
DSKY Loop Branches away to Interrupt Service Routine


REG
Register and Math
REGDATA
uSEQ Microinstruction Bits
ADD,SUB,CARRYEN
HOLD,SHL,SHR
SEL[X9,SUM,X,Y]
SEL[0,X]
LOAD XMSD
LOAD ZERO
0
X9
ENABLE, LOAD, CNTUP/DN
SEL[Dat,Q]
IMMDATA
XmsdIn
Cmd
X7
XREG
XlsdIn
X0
SHR X9
SHR ZERO/DIV 10
SHR SUM
SHR YLSD
ROR
SHL IMMDAT/MULT 10
SHL QMSD
XDig0[3:0]
DSKY
HOLD,SHL,SHR
SEL[0,Y,X]
YmsdIn
Cmd
Y7
NC
YREG
NC
YlsdIn
Y0
SHR XLSD
ROR
SHR ZERO/DIV 10
HOLD,SHL,SHR
SETBITS, NXTPH
QmsdIn
Cmd
Q7
QREG
QlsdIn
Q0
SHR YLSD
SHL ZERO/MULT 10
SHL QL
SEL[0,QL]
LOAD QL
LOAD ZERO
QL
ENABLE, LOAD, CNTUP/DN
Cmd
Oprnd2
ARITHEXP
Sum
Dp
XDp
Oprnd1
STATPHSRST
Operation
Tracking &
Status Bits
Bra Cond's
SEQ
Plumbed Data Flows
Key
Disp
Xreg
Yreg
Qreg
+
-
X,Y,QREG
msd
lsd
SHR
SHL

REG: Registers, Data Steering and Compute

- REG is a passive unit with no autonomy. Completely controlled by SEQ
- Three Registers; XReg, YReg and QReg are each arrays of 8 BCD (4-bit) digits
  - Each register supports shift operations; either SHR, SHL, or both. SHR moves digits one position from MSD (most significant digit) to LSD, and the current LSD exiting the register with SRC replacing the MSD. SHL moves digits from LSD to MSD and the MSD exiting the register with SRC replacing the LSD
    - SHRX SRC: X9, SUM, X_{LSD}, Y_{LSD}
    - SHLX SRC: IMMData, Q_{MSD}
    - SHRY SRC: Zero, Y_{LSD}, X_{LSD}
    - SHRQ SRC: Zero, Y_{LSD}
    - SHLQ SRC: Zero, Q_{LSD}
  - Each register may operate individually or in tandem with others but only limited cases of simultaneous register shifting are supported. Examples include: Exchange the values of Xreg and Yreg; Adding Xreg and Yreg, with SUM replacing Xreg while Yreg circulates (or replaced with zero); Copy Xreg to Yreg while Xreg circulates (or replaced with zero)
- X9 and QL are single BCD digit registers (loadable counters), auxiliary to Xreg and Qreg, respectively.
  - X9 accumulates carries out of Xmsd for Addition and Multiplication operations and is used to test Xreg digits for 0
  - QL is used for Multiplication as a loop counter and for Division as partial quotient accumulator
  - Each can increment or decrement by one and each is loadable from two sources:
    - X9 SRC: Zero, X_{MSD}
    - QL SRC: Zero, Q_{LSD}

ArithExp – BCD Adder Subtractor

- ArithExp is a single digit BCD adder/subtractor that operates only on X_{LSD} and Y_{LSD}. The output of Arith is the SUM routed only to X_{MSD}
  - Additional outputs indicate if the add generated a carry out and if the result of a subtraction is a negative number.
  - Subtraction is done through 10s complement arithmetic.
- The Add/Sub block is implemented with a cascade of three combinational logic stages:
  - 10's complement of one input (subtraction)
  - Summing the inputs
  - BCD correction of result
- The CarryO bit is valid for exactly one clock following the sum. This imposes a requirement for any operation on multi-digit numbers be carried out with contiguous add instructions (no intervening cycles)
- The IsNeg bit is valid for exactly one clock following the sum. This requires that action to be taken as a result of the IsNeg signal is immediate.
- If the result of a subtraction is negative, the 10's complement result is stored in Xreg and additional uCode performs subtraction of the result from 0, returning the desired result

ArithExp – Metadata Counters

- ArithExp includes 4 counters that react to key strokes to compute sum of exponents, difference of exponents, digit count, and decimal point position
  - Incremented or decremented by microcode instructions
    - XDigCnt – increments for each numeric key. If zero(s) is the first keystroke, it is not counted
    - XDpCnt – increments for each numeric key after the decimal point (counts digits in fractional portion of operand)
    - SexpCnt, DexpCnt – hold the sum and difference, respectively, of the exponents of the two operands
      - Positive Exponent - first key of operand is numeric. Increment for each numeric key until decimal point is keyed
      - Negative Exponent - first key of operand is decimal point key. Decrement for each leading zero keyed until non zero numeric key
      - SexpCnt
        - for each operand, positive exponents increment up, negative exponents decrement down
      - DexpCnt
        - for first operand, positive exponents increment up, negative exponents decrement down
        - For second operand, negative exponents increment up, positive exponents decrement down
- Via microcode instructions, DexpCnt can be transferred to SexpCnt. SexpCnt can be transferred to XDpCnt. DexpCnt can be transferred to XDpCnt in two steps

StatPhsRst

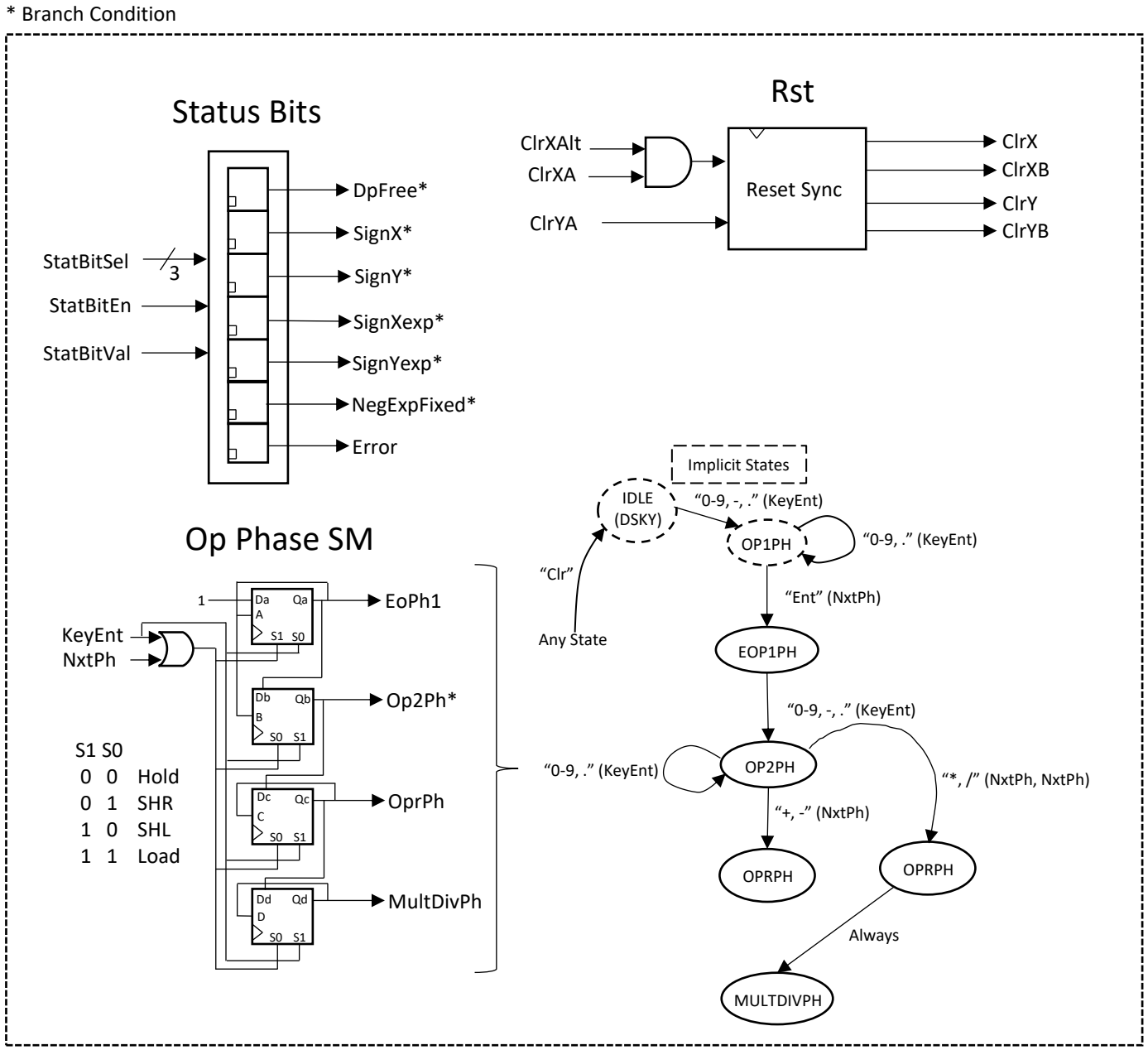
- There are two reset domains for sequential logic in REG, X and Y
  - The Xreg domain includes Xreg and state associated with Xreg
    - X reset is triggered by a microcode instruction and by the transition from entering the first operand to entering the second operand
  - The Yreg domain includes Yreg and Qreg and associated state
    - Y reset is triggered by a microcode instruction
  - The Rst part of StatPhsRst is a reset synchronizer to filter spurious logic glitches from SEQ on X and Y domain resets
  - It also includes single pulse generator that generates an alternate X domain reset when the first key of the second operand is detected
- The Stat part of StatPhsRst is an array of status bits to store sign bits for operands and exponents and other state information needed to parse keystrokes. Bits are set and cleared by microcode instructions and by reset events

StatPhsRst

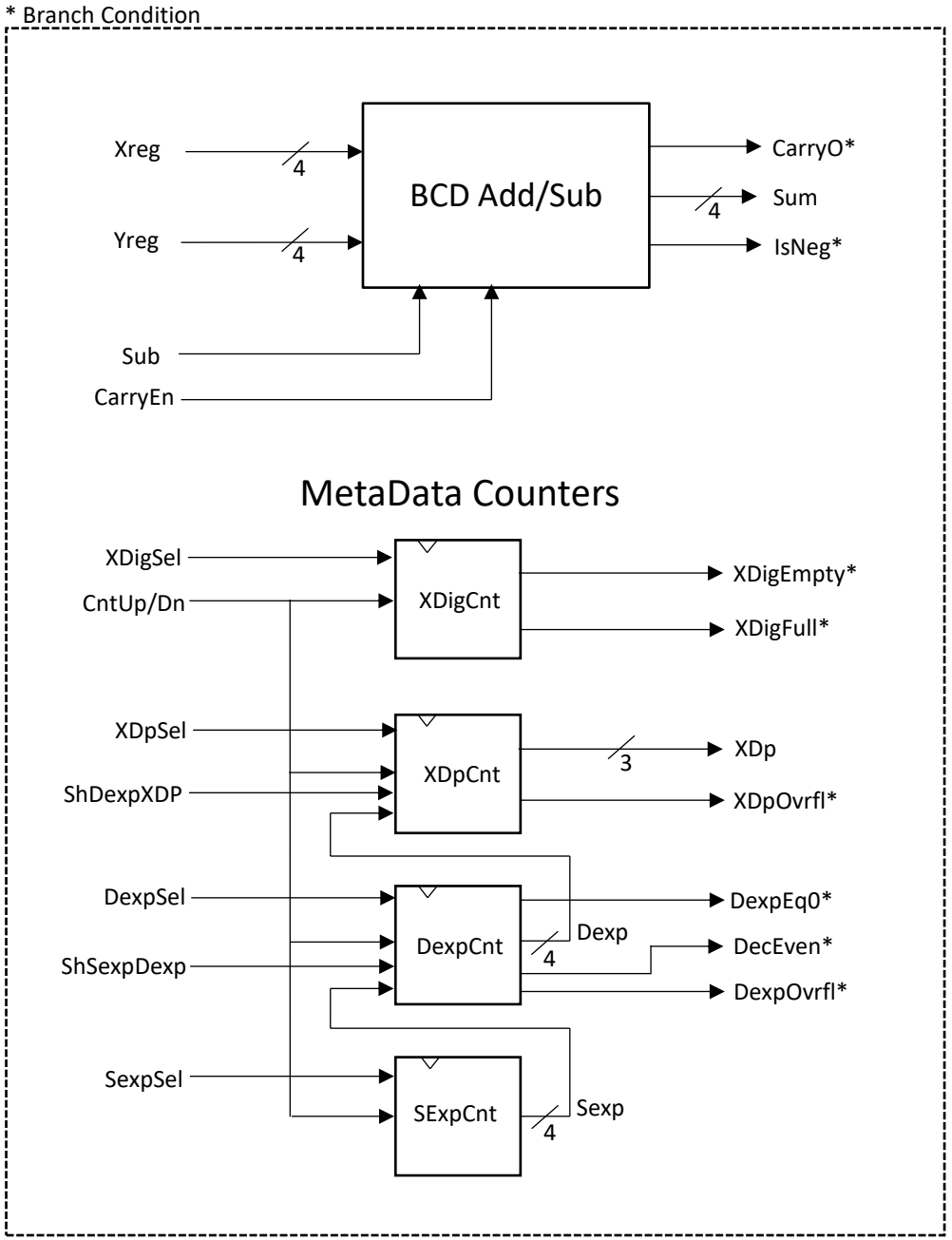
- The Phs part of StatPhsRst is a state machine that tracks the phase of the operation underway. Its state variables are tested by branch instructions
- States include
  - Idle* – System has cycle through reset but no keys have been pressed (virtual state)
  - Op1Ph* – First operand is being entered (virtual state)
  - EOp1Ph – First operand is entered but second operand has not started
  - Op2Ph – Second operand is being entered
  - OprPh – One of the 4 operations is underway

  * Idle state and Op1Ph state do not need actual state variables but they represent operational states of the overall machine
- The state machine is advanced by a microcode instruction
  - The transition from EndOp1 to Op2Ph will also happen with the first key starting operand2

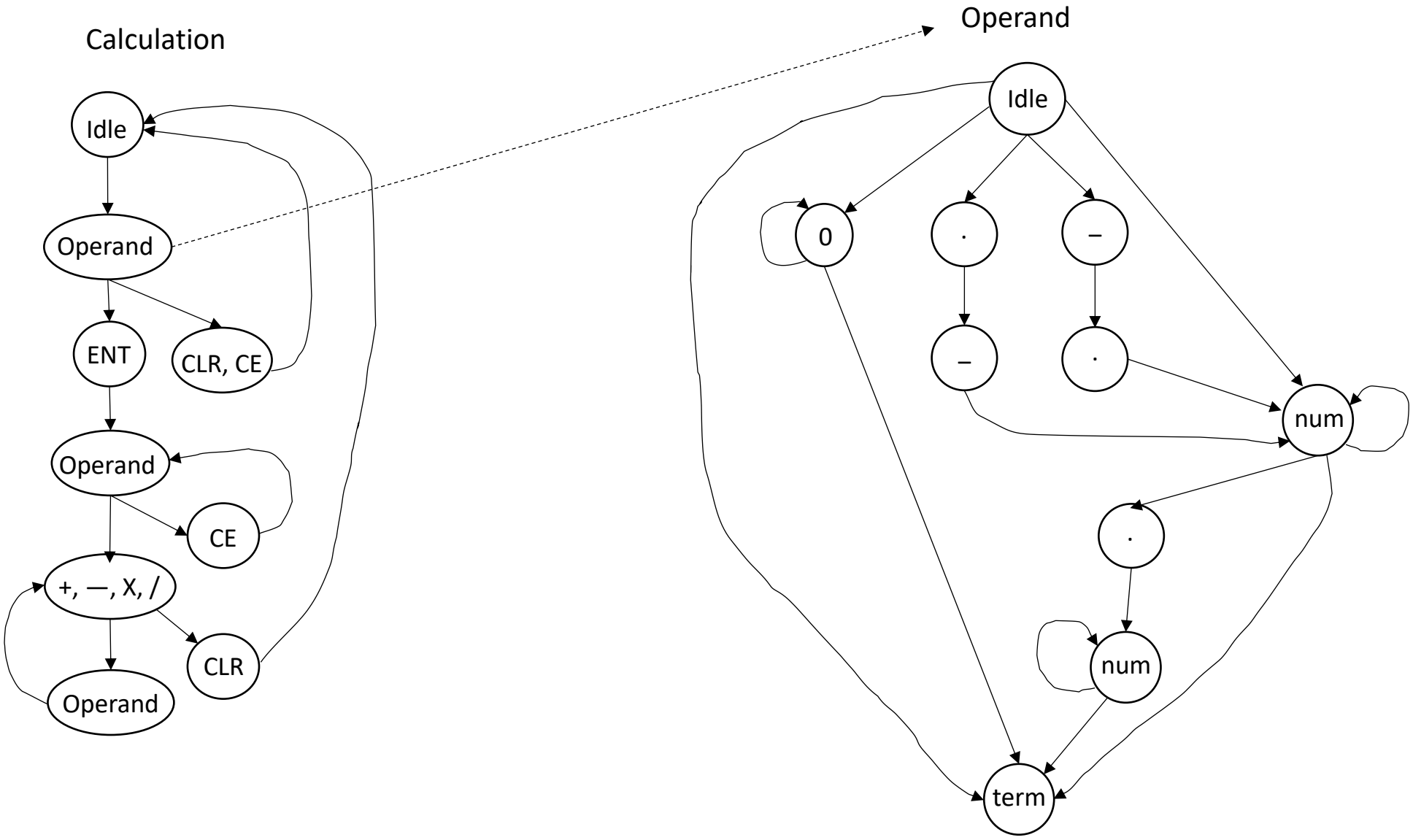
StatPhsRst


* Branch Condition
Status Bits
StatBitSel
3
StatBitEn
StatBitVal
DpFree*
SignX*
SignY*
SignXexp*
SignYexp*
NegExpFixed*
Error
Rst
ClrXAlt
ClrXA
ClrYA
Reset Sync
ClrX
ClrXB
ClrY
ClrYB
Op Phase SM
1
KeyEnt
NxtPh
EoPh1
Op2Ph*
OprPh
MultDivPh
S1 S0
0 0 Hold
0 1 SHR
1 0 SHL
1 1 Load
Implicit States
IDLE (DSKY)
"0-9, -, ." (KeyEnt)
OP1PH
"0-9, ." (KeyEnt)
"Clr"
Any State
"Ent" (NxtPh)
EOP1PH
"0-9, -, ." (KeyEnt)
"0-9, ." (KeyEnt)
OP2PH
"*, /" (NxtPh, NxtPh)
"+, -" (NxtPh)
OPRPH
OPRPH
Always
MULTDIVPH

ArithExp

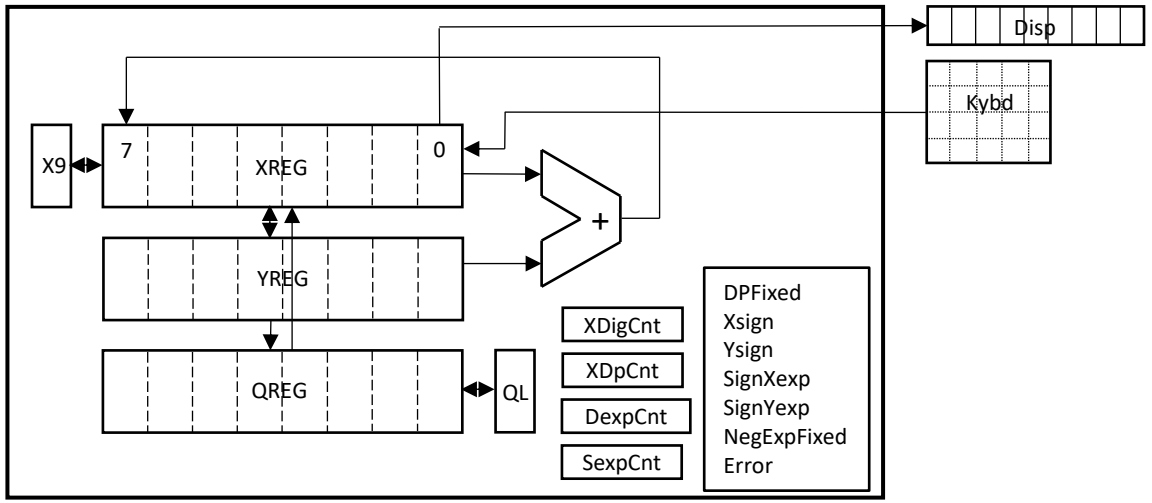

* Branch Condition
Xreg
4
Yreg
4
BCD Add/Sub
CarryO*
Sum
4
IsNeg*
Sub
CarryEn
MetaData Counters
XDigSel
CntUp/Dn
XDigCnt
XDigEmpty*
XDigFull*
XDpSel
ShDexpXDP
XDpCnt
3
XDp
XDpOvrfl*
DexpSel
ShSexpDexp
DexpCnt
4
Dexp
DexpEq0*
DecEven*
DexpOvrfl*
SexpSel
SExpCnt
4
Sexp

Key Sequence For Operations



How Operands Are Entered

REG Unit Data Model



XDigCnt – Counts numeric digits entered into Xreg
XDpCnt – Counts digits after decimal point in Xreg
DexpCnt – Computes difference between exponents in Xreg and Yreg
SexpCnt – Computes sim of exponents in Xreg and Yreg
NegExpFixed – Indicates that accounting for negative exponent is done

Key	Yreg 76543210	Xreg/Disp 7	6	5	4	3	2	1	0	Xdig	XDp	Sexp	Dexp	DpFix	NEF	PhsSt
num									1.	1	0	1	1	0	0	op1ph
num								1	2.	2	0	2	2	0	0	op1ph
Dp								1	2.	2	0	2	2	1	0	op1ph
num							1	2.	3	3	1	2	2	1	0	op1ph
num						1	2.	3	4	4	2	2	2	1	0	op1ph
Ent	12340000	1	2.	3	4	0	0	0	0	8	6	1	1	1	0	eop1ph
num	12340000								9.	1	0	2	0	0	0	op2ph
num	12340000							9	8.	2	0	3	-1	0	0	op2ph
num	12340000						9	8	7.	3	0	4	-2	0	0	op2ph
Dp	12340000						9	8	7.	3	0	4	-2	1	0	op2ph
num	12340000					9	8	7.	6	4	1	4	-2	1	0	op2ph
+	12340000	9	8	7.	6	0	0	0	0	8	5	3	-1	1	0	oprph
Dp									0.	1	0	0	0	1	0	op1ph
num								0.	0	2	1	-1	-1	1	0	op1ph
num							0.	0	0	3	2	-2	-2	1	0	op1ph
num						0.	0	0	0	4	3	-3	-3	1	0	op1ph
num					0.	0	0	0	1	5	4	-3	-3	1	1	op1ph
num				0.	0	0	0	1	2	6	5	-3	-3	1	1	op1ph
num			0.	0	0	0	1	2	3	7	6	-3	-3	1	1	op1ph
Ent	12300000	0.	0	0	0	1	2	3	0	8	7	-4	-4	1	1	eop1ph
num	12300000								9.	1	0	-3	-5	0	0	op2ph
Dp	12300000								9.	1	0	-3	-5	1	0	op2ph
num	12300000							9.	8	2	1	-3	-5	1	0	op2ph
num	12300000						9.	8	7	3	2	-3	-5	1	0	op2ph
+	12300000	9.	8	7	0	0	0	0	0	8	7	-4	-4	1	0	oprph

Dec	Bin
-14	0010
-13	0011
-12	0100
-11	0101
-10	0110
-9	0111
-8	1000
-7	1001
-6	1010
-5	1011
-4	1100
-3	1101
-2	1110
-1	1111
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110

MicroCode Sample

7.5.3			I[5:0], I23 1[5:0], 3[7]	2[7:0], 3[2:0]	I[3:0] 1[3:0]	I[22] 3[6]	I[13:8] 2[5:0]	I[7:5] 1[7:5]	I[18:16] 3[2:0]	I[21:19] 3[5:3]	I[3:0] 1[3:0]	I[14] 2[6]	I[15] 2[7]	Logisim uROM Image D[7:0] I[15:8] [23:16]			
Addr	**Hex**	**Label**	**SEQ**		**DSKY**		**REG**					**AddSub**		**1[7:0]**	**2[7:0]**	**3[7:0]**	**4**
			Bra	**Target**	**Sel[3:0]**	**KeyAck**	**RegCmd**	**SelCtl1**	**SelCtl2**	**SelCtl3**	**KeyMCntSel**	**Sub_AddB**	**CarryEn**	v2.0 raw			Required for Logisim
675	2A3	**ENTOP1**	BRSignXexp	SETOP1DPFIR										3E	A9	02	If Op1 has neg exponent (DPFirst) then save in SignYexp (
678	2A6		JMP	TSTDIGEMPTY										10	AC	02	Otherwise goto test for empty X reg
681	2A9	SETOP1DPFIR							StatBitEn	StatBit1	SignYexp			05	00	1B	
684	2AC	TSTDIGEMPTY	BRDigEmpty	ENTERZERO										32	B5	02	If no digit hit yet, Jump to enter0
687	2AF		BRDPFree	ADJEXPFIXDP										37	BE	02	If DP was never hit then goto adjust exp and fixDP
690	2B2		JMP	LEFTJUST										10	C7	02	Otherwise go to left justify
693	2B5	ENTERZERO							CntrEn	CntrIncr	XDigCnt			01	00	36	Incr Xreg Dig count (enter a 0)
696	2B8								StatBitEn	StatBit1	FixDP			00	00	1B	and fix the DP
699	2BB		JMP	LEFTJUST										10	C7	02	then left justify
702	2BE	ADJEXPFIXDP							CntrEn		XYSexpCnt			04	00	06	If X is whole integer, fix DP and adjust exponent
705	2C1								CntrEn		XYDexpCnt			03	00	06	
708	2C4								StatBitEn	StatBit1	FixDP			00	00	1B	
711	2C7	LEFTJUST	BRDigFull	COPYXY										31	D6	02	Left justify X until Xdig is full
714	2CA						LeftJustX	SelXData			0			00	02	00	Shift left with 0's in at msd
717	2CD								CntrEn	CntrIncr	XDigCnt			01	00	36	and adjust Xdp and Xdig count to match
720	2D0								CntrEn	CntrIncr	XDPCnt			02	00	36	
723	2D3		JMP	LEFTJUST										10	C7	02	
726	2D6	COPYXY				InitLp8								00	00	40	Copy Xreg to Yreg
729	2D9		BRLoop8NZ	COPYXYE										3F	DC	02	
732	2DC	COPYXYE					CopyXY	SelXReg	SelXXlsd	SelYXlsd				20	05	28	
735	2DF		BRLoop8NZ	COPYXYE										3F	DC	02	
738	2E2		BRSignX	CPSIGXSIGY										34	E8	02	Copy Xsign to Ysign
741	2E5		JMP	RMLDZEROS										10	EB	02	
744	2E8	CPSIGXSIGY							StatBitEn	StatBit1	SignY			04	00	1B	
747	2EB	RMLDZEROS				InitLp8								00	00	40	Loop to remove leading zeros from fractional args. Clea
750	2EE	BRN1E	BRLoop8NZ	RLZENTERE										3F	F4	02	Burn loop 1 of 9 and jump to entry point
753	2F1	RLZTOPE					RmLdZeroX	SelXData			0			00	02	00	Shift Xmsd out the top and zero in from the bottom

MicroCode Fields, Mnemonics and Codes

	Jmp/Bra	1[4]	1[3:0]
	JMP	1	0000
BR0	BRDecEven	1	0000
BR1	BRDigFull	1	0001
BR2	BRDigEmpty	1	0010
BR3	BRDintOvrfl	1	0011
BR4	BRSignX	1	0100
BR5	BRSignY	1	0101
BR6	BRInOp2Ph	1	0110
BR7	BRDPFree	1	0111
BR8	BRQLZero	1	1000
BR9	BRX9Zero	1	1001
BR10	BRDpOvrfl	1	1010
BR11	BRNoKeyDet	1	1011
BR12	BRIsNeg	1	1100
BR13	BRCarryO	1	1101
BR14	BRSignXexp	1	1110
BR15	BRLoop8NZ	1	1111
	BRSignYexp	1	0110
	BrNegExpTerm	1	0101
	ExtraBR2	0	0000
	ExtraBR3	0	0000
	HWINTEN	0	0000

RegCmd	2[5:0]
CircX	000001
DPAdjX	000001
EntNumX	000010
LeftJustX	000010
DPAdjY	000100
CopyXY	000101
ExchXY	000101
AddSub	000101
DivXY10	000101
ShrX9X8QDiv10	010001
CopyYQ	010100
CopyQX	100010
ApendDivdn	000010
RmLdZeroX	000010
CopyQ0QL	100000
ShrX	000001
ShlX	000010
CopyYX	000101
ExtraCmd3	000000

SelCtl1	1[7:5]
SelXReg	001
LoadX9	010
SelXQmsd	011
LoadQL	100
SHDexpXDP	101
Reserved1	110
Reserved2	111
HWINTEN	000
SelXData	000
SelXSumX9	000
ExtraSelCtl1	000
ExtraSelCtl2	000
ExtraSelCtl3	000
ExtraSelCtl4	000

BR*	001

SelCtl2	3[2:0]
ClrXPh1	001
ClrYPh2	010
StatBitEn	011
SelXYlsd	100
SelXSum	100
NxtPh	101
CntrEn	110
SHSexpDexp	111
SelXX9	000
SelXXlsd	000
ExtraSelCtl1	000
ExtraSelCtl2	000

ALUCmd	2[7:6]
SubXY	x1
CarryEn	1x

CtlSel3	3[5:3]
SelQLQlsd	001
SelYQYlsd	010
StatBit1	011
SelX9Xmsd	100
SelYXlsd	101
CntrIncr	110
RowHold	111
SelYZero	000
SelQZero	000
SelX9Zero	000
SelQLZero	000
ExtraSelCtl3	000

SEQ/DSKY	3[7:6]
HWINTEN	10
KeyAck	01
InitLP8	01

KeyMCntSel	1[3:0]
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
BlankNum	1111
XDigCnt	0001
XDPCnt	0010
XYDexpCnt	0011
XYSexpCnt	0100
X9Cnt	0101
QLCnt	0110
FixDP	0000
SignXexp	0001
NegExpTerm	0010
SignX	0011
SignY	0100
SignYexp	0101
AssertError	0110

ImmData asserts on Ph2 of each cycle

SelCtl1, SelCtl2, RegCmd, ALUCmd assert on Ph4 and is valid only Ph4

SelCtl3, KeyAckB, HWIntEn Assert on Ph4 of each cycle

SelCtl3 Can assert simutaneous with JMP/BR*. SelCtl1/2 cannot.

MicroInstruction Fields

SelCtl1 (I5br:I7)

SelXReg	001	Select X reg MSD source from X reg or Y reg group
LoadX9	010	Load X9 source into X9 reg
SelXQmsd	011	Select X reg source from Q Dig7
LoadQL	100	Load QL source into QL reg
SHDexpXDP	101	Load Dexp counter value into XDP counter
Reserved1	110	Currently unassigned
Reserved2	111	Currently unassigned
SelXData	000	Select X LSD reg source from D[3:0]
SelXSumX9	000	Select X reg MSD source from Sum or X9 group

SelCtl2 (I16:I18)

SelQLQlsd	001	Select QL reg source from Q reg LSD
SelYQYlsd	010	Select Y and Q reg MSD sources from Y reg LSD
StatBit1	011	Set selected status bit to 1
SelX9Xmsd	100	Select X9 reg source from X reg MSD
SelYXlsd	101	Select Y reg source from X reg LSD
CntrIncr	110	Increment selected and enabled counter
RowHold	111	DSKY row latch disable (hold)
SelYZero	000	Select Y reg MSD source from 0
SelQZero	000	Select Q reg MSD source from 0
SelX9Zero	000	Select X9 reg source from 0
SelQLZero	000	Select QL reg source from 0

SelCtl3 (I19:I21)

ClrXPh1	001	Clear X reg domain devices
ClrYPh2	010	Clear Y reg domain devices
StatBitEn	011	Enable status latches for write
SelXYlsd	100	Select X reg MSD source from Y reg LSD
SelXSum	100	Select X reg MSD source from SUM
NxtPh	101	Advance tracking state machine state
CntrEn	110	Enable selected counter for up/down advance
SHSexpDexp	111	Load Sexp counter value into Dexp
SelXX9	000	Select X reg MSD source from X9 reg
SelXXlsd	000	Select X reg MSD source from X reg LSD

RegCmd (I8:I13)

CircX	000001	Shift X reg from MSD toward LSD
DPAdjX	000001	
ShrX	000001	
EntNumX	000010	Shift X reg from LSD toward MSD
LeftJustX	000010	
ApendDivdn	000010	
RmLdZeroX	000010	
DPAdjY	000100	Shift X reg from MSD toward LSD
CopyXY	000101	Shift X and Y reg from MSD toward LSD
ExchXY	000101	
AddSub	000101	
DivXY10	000101	
ShrX9X8QDiv10	010001	Shift X and Q reg from MSD toward LSD
CopyYQ	010100	Shift Y and Q reg from MSD toward LSD
CopyQ0QL	100000	Shift Q reg from LSD toward MSD
CopyQX	100010	Shift X and Q reg from LSD toward MSD

Branches (D3:D0)

JMP	NA	Unconditional Branch
BRDecEven	0000	Branch if DexpCnt=0
BRDigFull	0001	Branch of DigCnt=8
BRDigEmpty	0010	Branch if DigCnt=0
BRDintOvrfl	0011	Branch if Dexp>=8
BRSignX	0100	Branch if SignX status bit set
BRSignY	0101	Branch if SignY status bit set
BRInOp2Ph	0110	Branch if in or beyond Op2Ph tracking state
BRDPFree	0111	Branch of DP status bit has not been set
BRQLZero	1000	Branch if Qlreg=0
BRX9Zero	1001	Branch if X9reg=0
BRDpOvrfl	1010	Branch if XDpCnt>=8
BRNoKeyDet	1011	Branch if no key is currently pressed
BRIsNeg	1100	Branch if arithmetic (SUM) sign bit is set
BRCarryO	1101	Branch if arithmetic carry out bit is set
BRSignXexp	1110	Branch if Op2's exponent sign status bit is set
BRLoop8NZ	1111	Branch if LoopCnt>0
BRSignYexp	0110	Branch if Op1's exponent sign status bit is set
BrNegExpTerm	0101	Branch if current negative exponent calculated

Immediate Data Labels (D3:D0)

0	0000	Immedate data 0
1	0001	Immedate data 1
2	0010	Immedate data 2
3	0011	Immedate data 3
4	0100	Immedate data 4
5	0101	Immedate data 5
6	0110	Immedate data 6
7	0111	Immedate data 7
8	1000	Immedate data 8
9	1001	Immedate data 9
XDigCnt	0001	Select the XDigit counter
XDPCnt	0010	Select the XDP counter
XYDexpCnt	0011	Select the Dexp counter
XYSexpCnt	0100	Select the Sexp counter
X9Cnt	0101	Select the X9 counter
QLCnt	0110	Select the QL counter
FixDP	0000	Select the DP status bit
SignXexp	0001	Select the Op2 exponent sign status bit
NegExpTerm	0010	Select the negative exponent accumulation terminated status bit
SignX	0011	Select the Op2 sign status bit
SignY	0100	Select the Op1 sign status bit
SignYexp	0101	Select the Op1 exponent sign status bit
AssertError	0110	Select the Error status bit

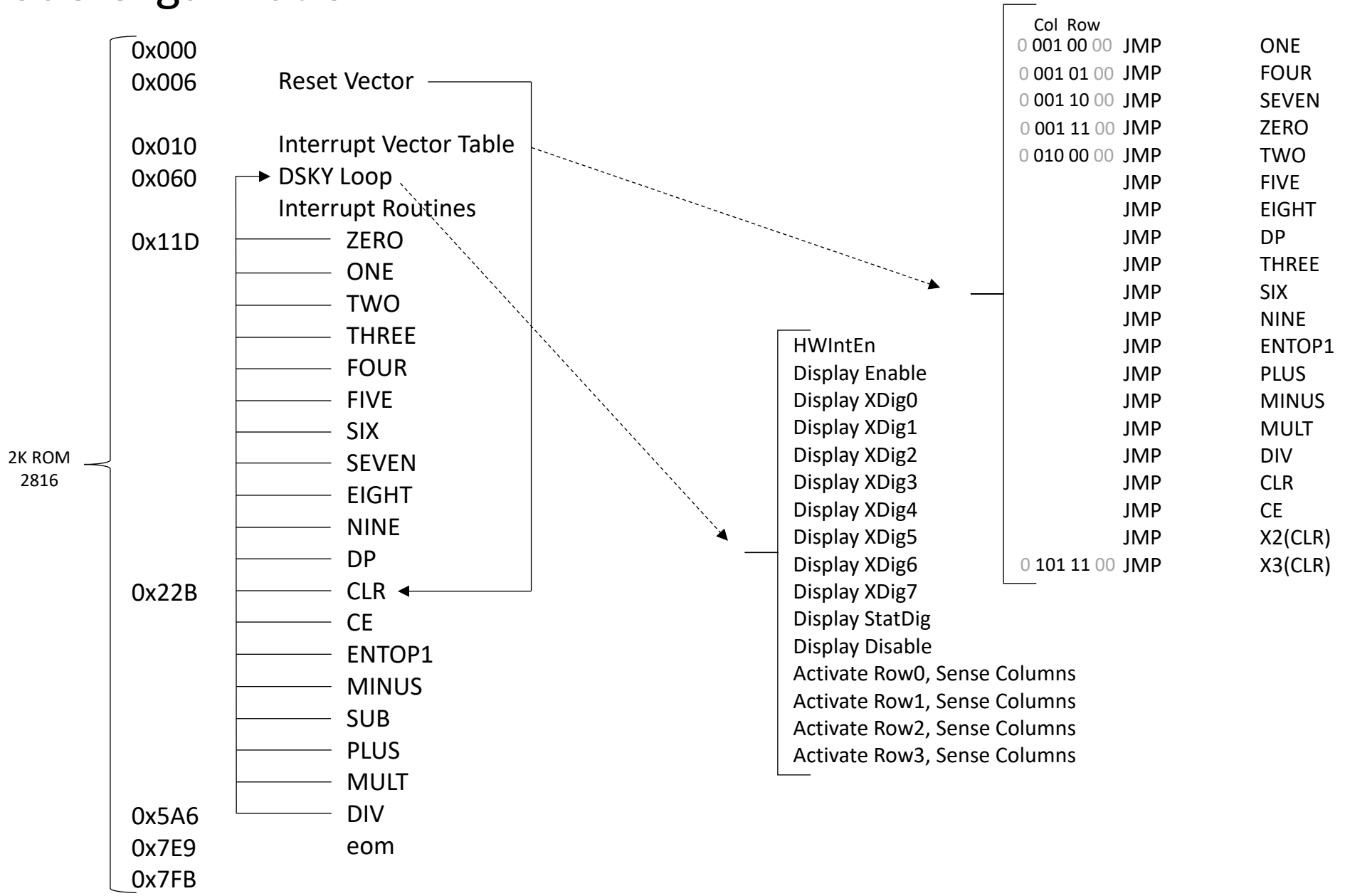
Add/Sub (I14:I15)

SubAddB	0	Add Digit 0 of Y register with Digit 0 of X register
	1	Subtract Digit 0 of X register from Digit 0 of Y register
CarryEn	0	Disable Carry In to Adder/Subtractor for one instruction
	1	Enable Carry In to Adder/Subtractor for one instruction

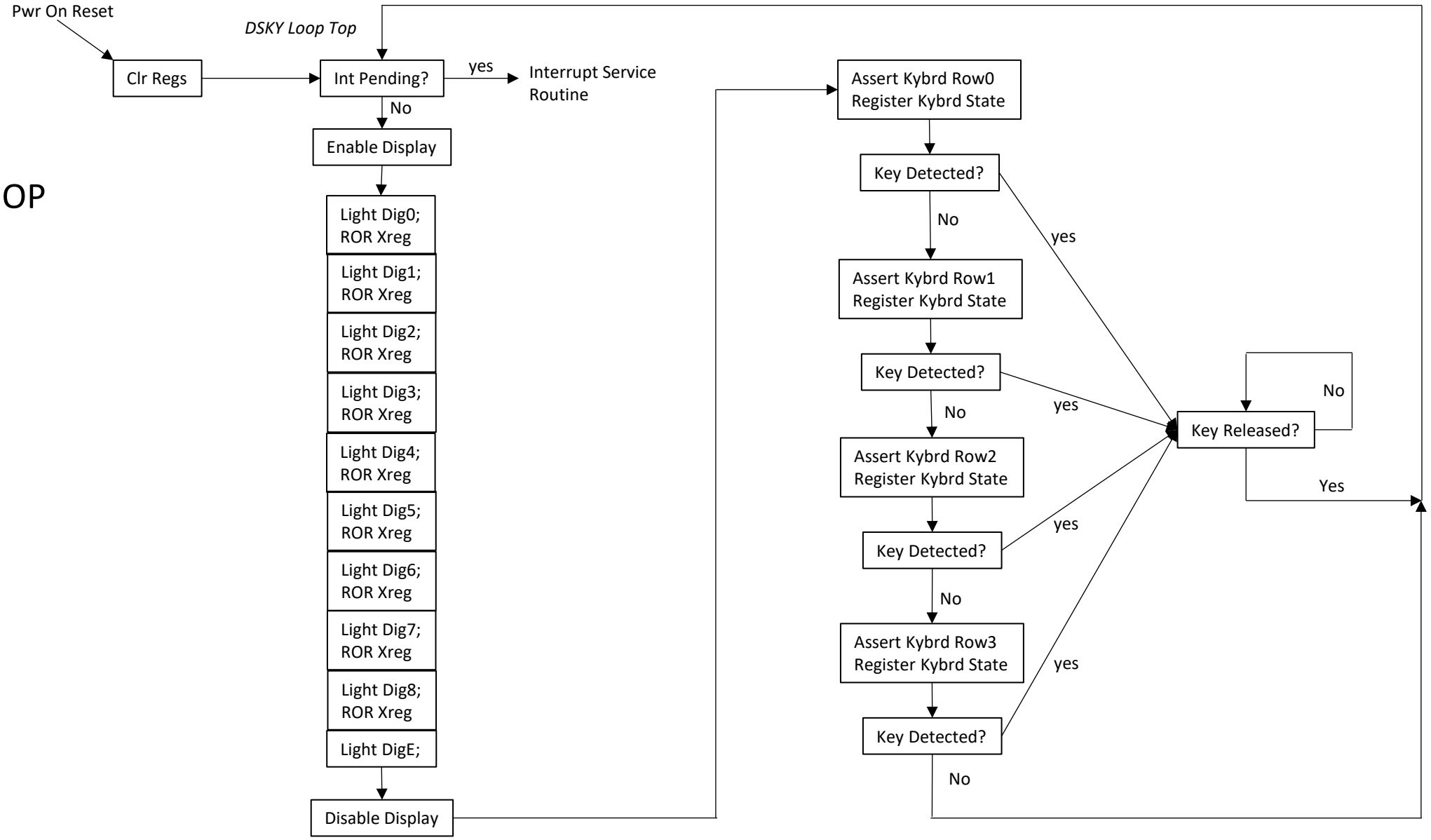
SEQ/DSKY (I22:I23)

KeyAck	0	NA
	1	Write state of keyboard into keyboard register, reset sequencer loop counter
HWIntEn	0	NA
	1	Enable branch to interrupt service routine if HWInt is active for one instruction

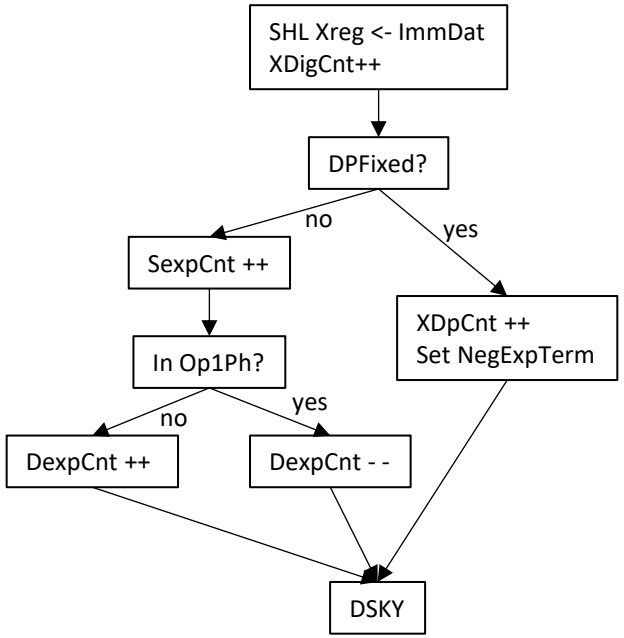
MicroCode Organization



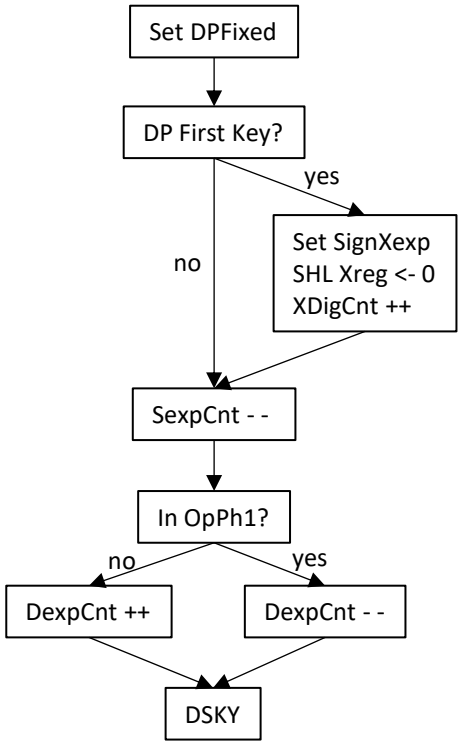
DSKY LOOP


Pwr On Reset
Clr Regs
DSKY Loop Top
Int Pending?
yes
Interrupt Service Routine
No
Enable Display
Light Dig0; ROR Xreg
Light Dig1; ROR Xreg
Light Dig2; ROR Xreg
Light Dig3; ROR Xreg
Light Dig4; ROR Xreg
Light Dig5; ROR Xreg
Light Dig6; ROR Xreg
Light Dig7; ROR Xreg
Light Dig8; ROR Xreg
Light DigE;
Disable Display
Assert Kybrd Row0 Register Kybrd State
Key Detected?
No
yes
Assert Kybrd Row1 Register Kybrd State
Key Detected?
No
yes
Assert Kybrd Row2 Register Kybrd State
Key Detected?
No
yes
Assert Kybrd Row3 Register Kybrd State
Key Detected?
No
yes
Key Released?
No
Yes

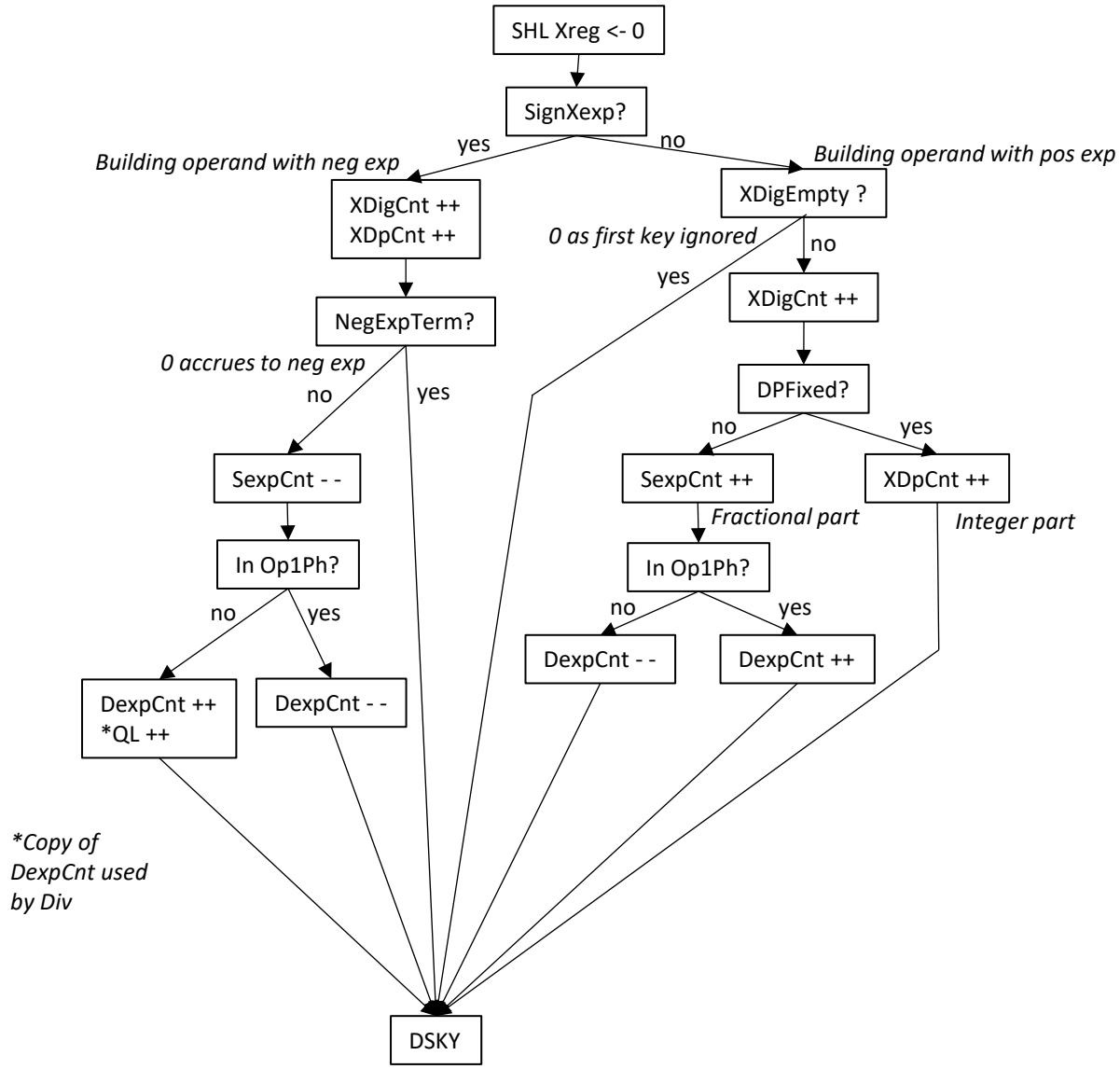
1-9 Service Routine



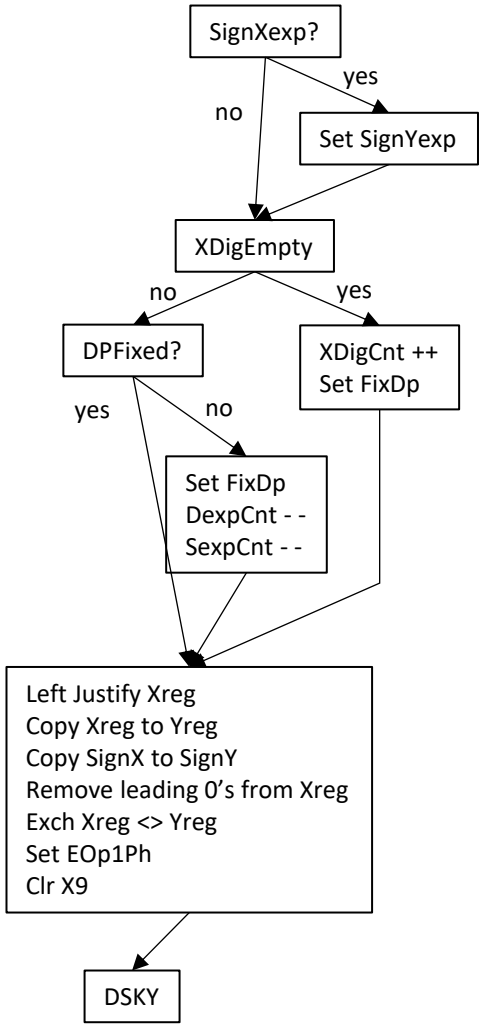
Dp Service Routine



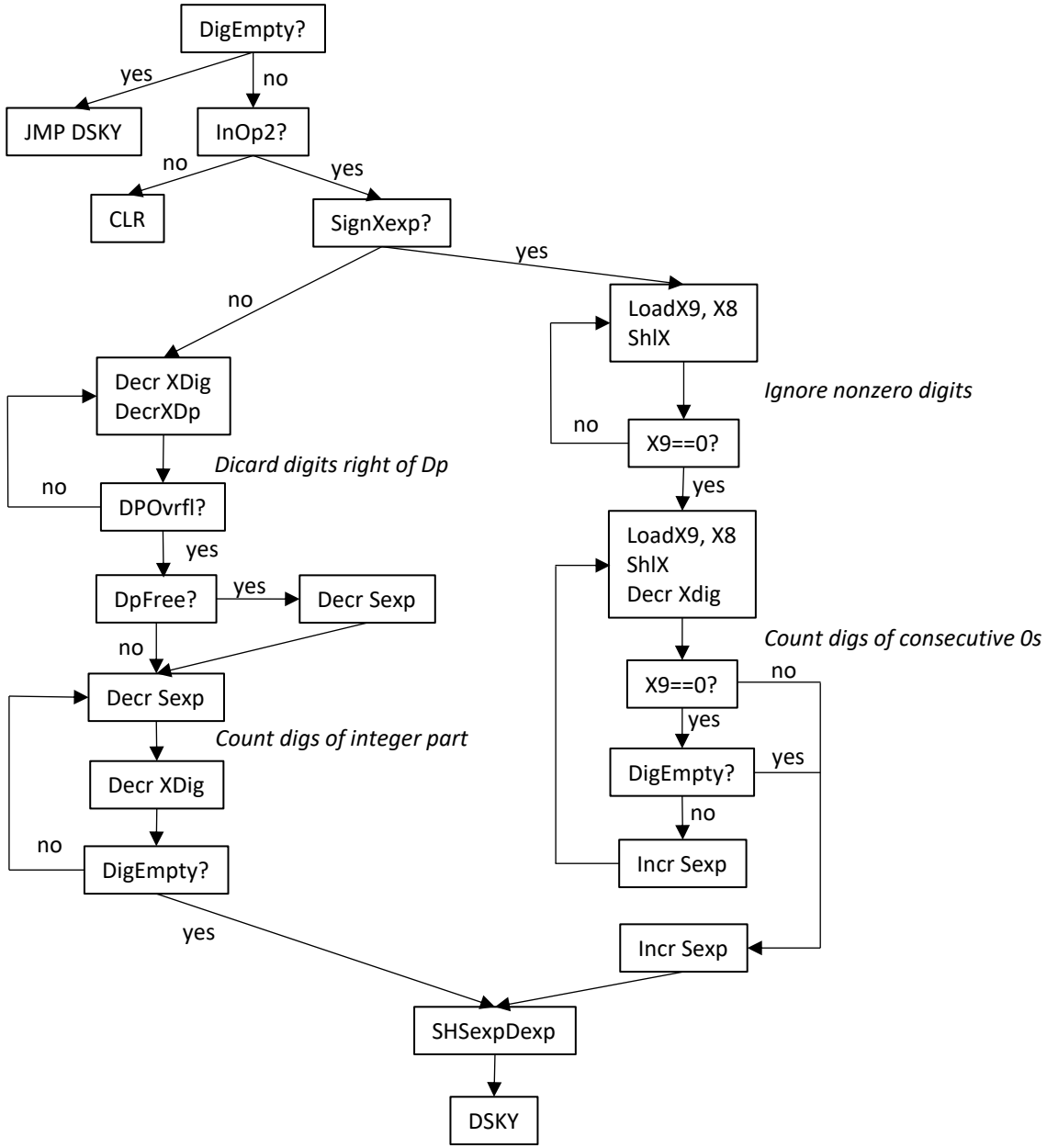
Zero Service Routine



Enter Op1 Service Routine



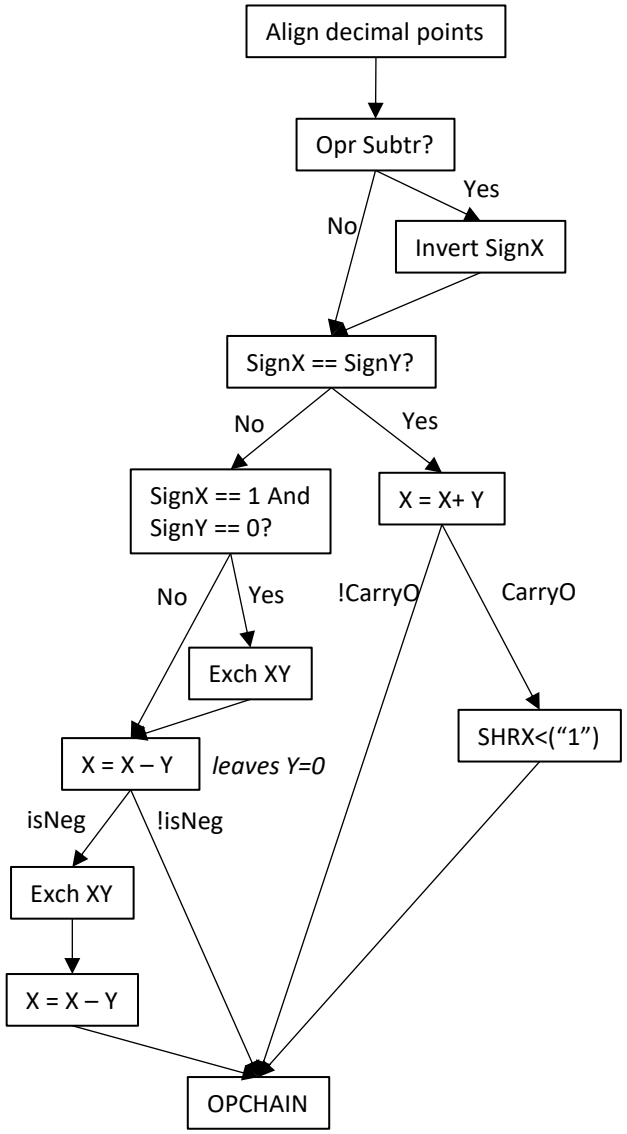
CE Service Routine



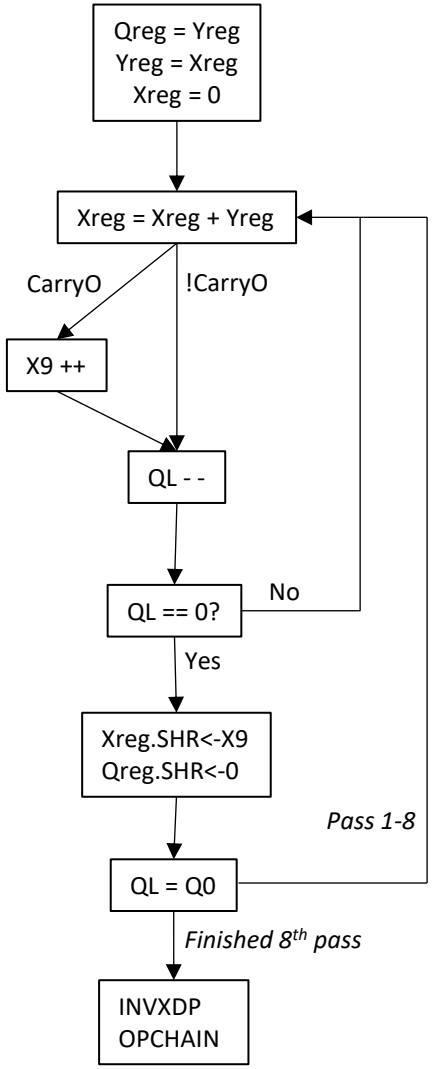
Basic Register Model

Xreg : Array[8:1] of BCD digits
X9 : BCD digit
Yreg : Array[8:1] of BCD digits
Qreg : Array[8:1] of BCD digits
QL : BCD digit
Xdp : Integer[2:0]
DpFixed : Boolean
Sexp : Integer[3:0]
Dexp : Integer[3:0]
NegExpDPFixed : Boolean

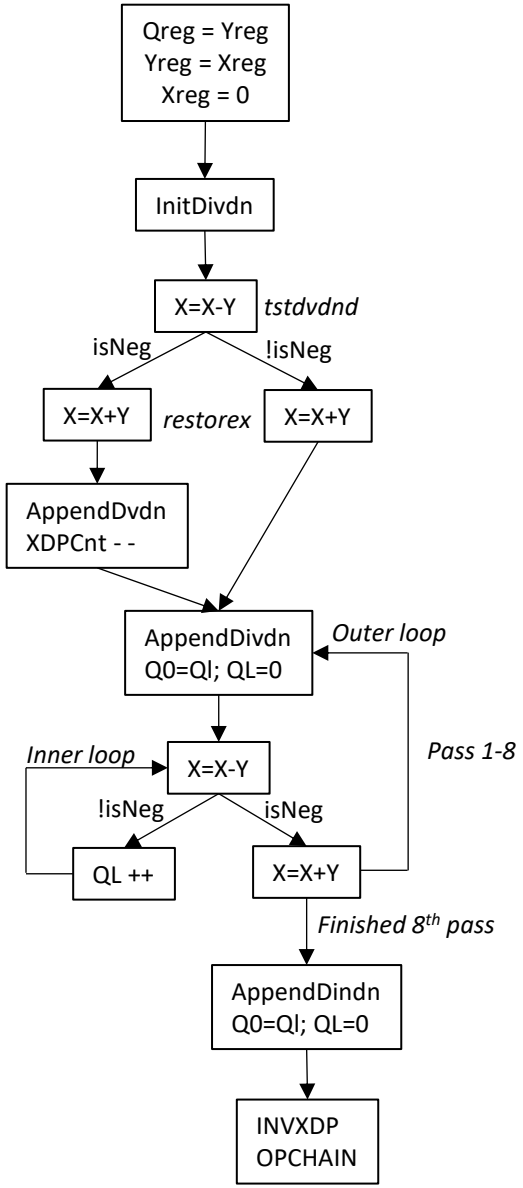
Add or Sub



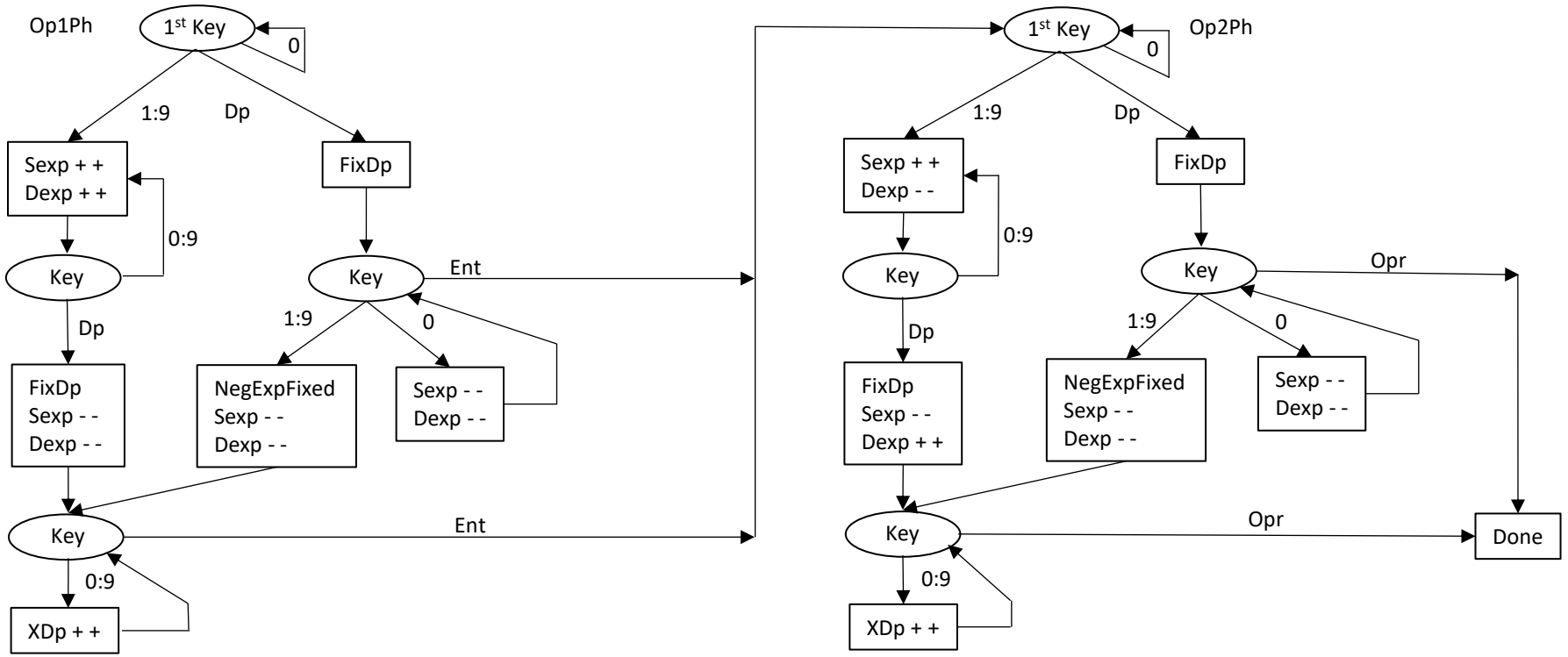
Mult



Div



Exponent Computation (XDp Placement)

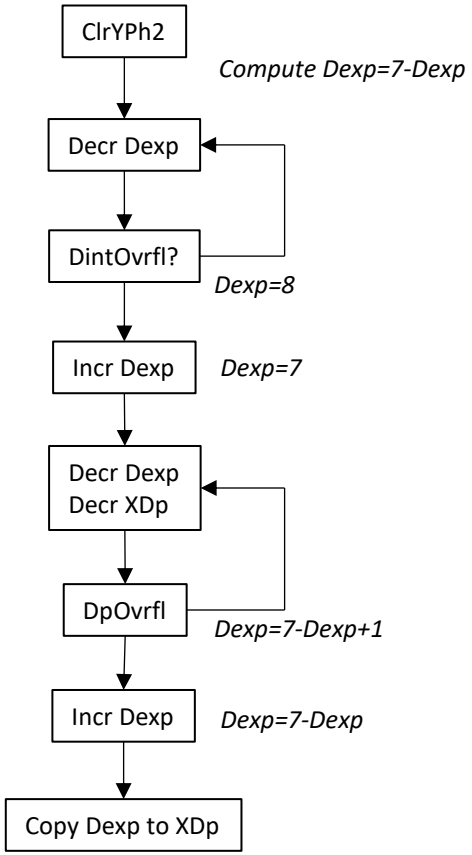


Add/Sub:
If Dexp <= 0, then XDp = XDp(Op2) else XDp = XDp(Op2) + Dexp.

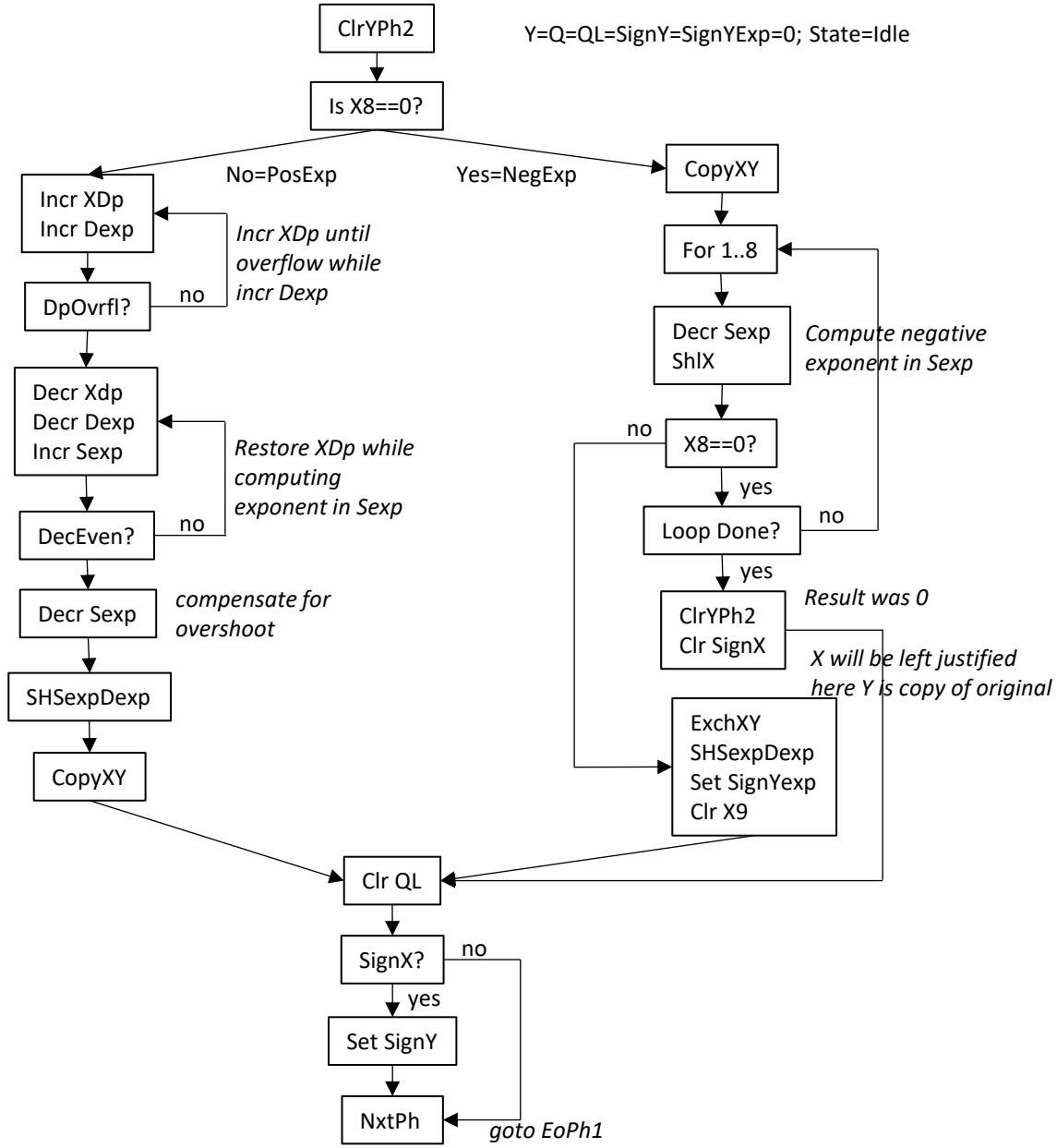
Mult
Pos Exp: XDp placement is initialized by Sexp. Incremented once (XDp shifted right one digit) if the final Mult loop leaves a non-zero accumulation of CarryOs in X9.
Neg Exp: XDp placed in left most position. Sexp number of leading zeros right shifted into MSD of product, losing right most digits for each shift.

Div:
Pos Exp: XDp placement is initialized by Dexp. Decremented once (XDp shifted left one digit) if the minimum number of dividend digits needed to subtract the divisor with a positive result is one more than the divisor digits.
NegExp: XDp placed in left most position. Dexp number of leading zeros right shifted into MSD, losing rightmost digits for each shift.

Invert XDP



Operation Chaining (Post processing the result of an operation to support chaining)



DSKY Loop

```
Do Forever
   If HWInt Then
      JMP IntTbl[IV]      // Execute key routine
   Enable Disp
   For (Digit 0 to 7)     // Light each digit
      Select Digit
      ROR Xreg
   Sel StatDigit
   Disable Disp
   For (Row 0 to 3)       // Scan keyboard
      If DetectKey
         Wait Until KeyReleased
```

1-9 Service Routine

```
SHL <num> into Xreg
Incr XdigCnt
If Dp_Not_Fixed then
   Incr SexpCnt
   If Op1Ph then
      Incr DexpCnt
   Else
      Decr DexpCnt
Else           // Dp Fixed.  Key hits no longer
               // accumulate exponents
   Incr XDpCnt
   Set NegExpTerm  // Indicate negative exponent
                   //accumulation terminated
```

Zero Service Routine

```
SHL “0” int Xreg
If Not SignXexp then    // If Xreg is > 1
   If XDigEmpty then    // And Xreg is empty
      JMP DSKY          // Then key hit can be ignored
   Else                 // Building operand > 1
      Incr XdigCnt
      If DpFixed Then   // Building the fractional part
         Incr XDp Cnt
         JMP DSKY
      Else              // Building the integer part
         Incr SexpCnt
         If Op1Ph Then
            Incr DexpCnt
         Else
            Decr DexpCnt
         JMP DSKY
Else                    // Building operand < 1
   Incr XdigCnt
   Incr XDpCnt
   If NegExpTerm Then   //  If no more leading 0’s then done
      JMP DSKY
   Else                 // Account for the larger negative exponent
      Decr SexpCnt
   If Op1Ph Then
      Decr DexpCnt
   Else
      Incr DexpCnt
      Incr QL       // QL used to mirror DexpCnt count
                    // of leading 0’s (used by Div)
   JMP DSKY
```

Dp Service Routine

```
Set FixDp
If Dp is first key hit then
   Set SignXexp
   SHL “0” into Xreg
   Incr XDigCnt
Decr SexpCnt
If Op1Ph then
   Decr DexpCnt
Else
   Incr DexpCnt
JMP DSKY
```

Enter Op1 Service Routine

```
If Dp was first key hit then  // Xreg < than 1
   Set SignYexp bit
If no numeric key was hit then
   Incr XDigCnt;  Set FixDp
Else
   If Dp was not hit then    // Xreg is pure integer
      Ser FixDp;  Dexp - -; Sexp - -
While not XDigFull           // Left justify Xreg
   SHL Xreg <- 0; XdigCnt ++; XDPCnt ++
Copy_Xreg_into_Yreg
Copy SignX into SignY
X9 <-  Xmsd       // Remove leading zeros.  Does nothing
                  //unless operand is < 1.
While X9 == 0               // The operand in X is mantissa
   X9 <- SHL Xreg <- 0      // The operand in Y is displayable version
Exchange Xreg<>Yreg
NxtPh             // Move to EOp1Ph
Clear X9
JMP DSKY
```

Add and Subtract Service Routine

```
MINUS
If Not Op2Ph Then     // If not in Op2 then "-" means sign bit
   Set SignX
   JMP DSKY
Else                  // If in Op2 then "-" means subtract
   Invert_SignX
PLUS
If DigEmpty Then      // If no key hit for Op2 then enter "0"
   Incr XDigCnt
   Set FixDp
Else
   If DpFree Then     // Op2 is pure integer, fix Dp and adj exp
      Set FixDp
      Decr SexpCnt
      Incr DexpCnt
Left_Justify_Xreg
While X8 == 0         //  Remove leading 0's from Op2<1
   SHL Xreg, 0
   Incr XDpCnt
While DexpCnt != 0    //  Get operands exponent-aligned (DpAdj)
   If DintOvrfl Then  // Shift Yreg right, count Dexp up
      SHR Yreg, 0
      Incr DexpCnt
   Else               // Shift Xreg right, count Dexp down
      SHR Xreg, 0
      Decr DexpCnt
While DpOvrfl         //  Restore leading 0's to Op2, if any
   SHR Xreg; SHR Yreg
   Decr XDpCnt
If (SignX XOR SignY) == 0 Then   // If Op signs same, then ADD
   Xreg = Xreg + Yreg
   If CarryO Then     // If final carry out then shift in X9, adj XDp
      Incr X9
      SHR Xreg, X9
      Decr XDpCnt
Else                  // Op signs are different
   If (Sign X == 1 AND SignY == 0) Then   // Yreg must be the negative op for ALU
      Exch Xreg<>Yreg
      Exch SignX<>SignY
   Xreg = Xreg – Yreg
If DpOvrfl Then
   Set Error
JMP DSKY
```

OPADD (Xreg + Yreg)

```
SHR Xreg, Sum; SHR Yreg, 0; ADD              // Shift Xreg and Yreg right,
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn     // with Lsd's going through ALU
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn     // and the Sum shifting into
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn     // Xreg msd, and 0's shifting
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn     // into Yreg msd
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; ADD, CarryEn
```

OPSUB (Xreg – Yreg)

```
SHR Xreg, Sum; SHR Yreg, 0; SUB
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
If IsNeg Then         //  Get 10's complement and BCD adjust
   Exch Xreg<>Yreg
   SHR Xreg, Sum; SHR Yreg, 0; SUB
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   SHR Xreg, Sum; SHR Yreg, 0; SUB, CarryEn
   Set SignX          // Since result is negative
While X8 == 0 AND NOT DpOvrfl    // Remove excess leading 0's, if any
   SHL Xreg, 0
   Incr XDpCnt
```

Mult Service Routine

```
If DigEmpty Then      //  If no key hit for Op2 then enter "0"
   Incr XDigCnt
   Set FixDp
Else
   If DPFree Then    // Else if Dp not hit then FixDp and adj exps
      Set FixDp
      Decr SexpCnt
      Incr DexpCnt
Left_Justify_Xreg
While X8 == 0       // Remove any leading 0's in Op2
   SHL Xreg, 0; Incr XDpCnt
NxtPh                // Move to OprPh
If (SignX XOR SignY) == 1 Then
   Set SignX
Else
  Clear SignX
Copy Yreg to Qreg   // Get regs set up for Mult loop
Copy Xreg to Yreg, clearing Xreg
Mult_Loop ----------
If (SignXexp == SignYexp == 0 AND DpOvrfl) Then   // Overflow handler
   Set Error
   JMP DSKY
If (SignXexp == SignYexp == 1 AND DpOvrfl) Then   // Underflow handler
   ClrXPh
   JMP DSKY
If DpOvrfl Then     // Product is <1
   While DpOvrfl    // Insert leading 0's to product <1
      SHR Xreg, 0    //   leaving XDP at 0 (7 after inv)
      Incr XDpCnt
Else                // Product is >1
   While X8 == 0    //  Remove leading 0's to product >1, if any
      SHL Xreg, 0
      Decr XDpCnt   // Move Dp to the left
Invert XDpCnt
JMP DSKY
```

```
Copy Q0 into QL; SHR Qreg
NxtPh                          // Move to MultDivPh
Shift SexpCnt into XDpCnt      // SexpCnt has the DP placement for product
For 8 Loops
   While QL != 0
      SHR Xreg, Sum; SHR Yreg, Y0; ADD             // Shift Xreg and Yreg right,
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn    //  with Xmsd getting getting the Sum
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn    //  and Ymsd circulating Ylsd
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn
      SHR Xreg, Sum; SHR Yreg, Y0; ADD, CarryEn
      If CarryO Then
         Incr X9
      Decr QL
   SHR Xreg, X9; SHR Qreg, 0
   Clr X9
If X9 != 0 Then
      SHR Xreg, X9
      Incr XDpCnt
```

Div Service Routine

```
        For 8  Loops                // Check for Op2==0
          Load X8 into X9
          If X9 == 0 Then
             SHR Xreg, X1
          Else Continue DPCHK
        Set Error
        JMP DSKY
DPCHK   If NOT DpFixed Then     // If Dp never entered
          Set FixDp                 //   Fix Dp and exp adj
          Decr SexpCnt
          Incr DexpCnt
          NxtPh                     // Move to OprPh
        If (SignX XOR SignY) == 0 Then    // Compute sign of quotient
          Clear SignX
        Else
          Set SignX
        If Op2 < 1 Then           // Divide both ops by 10 if Xreg is full
          If DigFull Then
             SHR Xreg, 0; SHR Y, 0
        Copy Yreg to Qreg
        Copy Xreg to Yreg, clearing Xreg
        Move to MultDivPh
        Copy DexpCnt to XDpCnt   // Exponent for quotient is DexpCnt
        If SignXexp Then       // Remove the extra "0" in front of Dp
          Decr XDigCnt
          While QL != 0
             Decr QL        // QL is a copy of DexpCnt to use for removing leading 0's
             Decr XDigCnt
        While NOT DigEmpty   // Init dividend with XDigCnt digits of Qreg
          SHL Xreg, Q8; SHL Qreg, 0
          Decr XDigCnt
        X = X-Y                   // Begin test if one more dividend digit is needed
        If IsNeg Then
          X = X+Y    // Restore dividend
          SHL Xreg, Q8; SHL Qreg, 0     // Add one more digit to dividend
          Decr XDpCnt
        Else
          X = X+Y    // Restore dividend
```

```
For 8 Loops    // outer loop
  SHL Xreg, Q8            // Append dividend with additional digit
  SHL Qreg, 0; Copy QL to Q0     // Accumulate quotient
  Clr QL
  While 1       // Inner loop
    X = X – Y
    If isNeg Break
    Incr QL
  X = X+Y
SHL Qreg, 0; Copy QL to Q0
Clr QL
Copy Qreg to Xreg          // Copy quotient to Xreg
If (SignYexp == 1 AND SignXexp == 0) Then   // Underflow handler
  If DPOvrfl Then
    Clr Xreg
    JMP DSKY
If (SignXexp == 1 AND SignYexp) == 0 Then  // Overflow Handler
  If DPOvrfl Then
    Set Error
    JMP DSKY
If NOT DPOvrfl Then   // Quotient >=1;  Done
  JMP DSKY
Repeat
  SHR Xreg, 0
  Incr XDpCnt
Until DPOvrfl
Invert XDpCnt
JMP DSKY
```

CE Service Routine

```
If Xreg is empty then
   JMP DSKY
If in Op1Ph then
   JMP CLR
Else
   If SignXexp then       // Back out the negative exponent
      While Xmsd != 0     //  Shift out leading Fs
         SHL Xreg <- 0
      While (Xmsd == 0)  AND (not XDigEmpty)   //  Count
         XDigCnt - -          // leading 0 digits and
         XYSexpCnt ++     // back out Op2 exponent
      XYSexpCnt ++
   Else                  // Back out the positive exponent
      Repeat             // Remove fractional part of operand from XDigCnt
         XDpCnt - -
         XDigCnt - -
      Until (DpOvrfl)
      If not DpFixed then
         XYSexpCnt - -           // Back out exp correction integer part
      While (not DigEmpty)    // Back out Op2 exponent
         XYSexpCnt - -
         XDigCnt - -
 Copy Sexp to Dexp
 ClrXPh1
Initialize Xreg with “FFFFFFF0”
JMP DSKY
```

CLR Service Routine

```
JMP RESET
```

Invert XDp

```
ClrYPh2
While not DintOvrfl
   XYDexpCnt ++
XYDexpCnt - -
While not DpOvrfl
   XYDexpCnt - -
   XYSexpCnt - -
XYDexpCnt ++
Copy Dexp to XDp
```

Operation Chaining

```
ClrYPh2; Clr QL    // Dexp and all other y regs are 0
X9 <- Xmsd
If X9 == 0 then      // Negative exponent
   Copy Xreg to Yreg
   For 1 to 8       //  Compute negative exponent
      XYSexpCnt - -  // decrement Sexp
      SHL Xreg          // and shift left X
      X9 <- Xmsd       // for every leading 0
      If X9 != 0 then   // Done, X is left just mantissa
         Exch Xreg <> Yreg  // X is Leftjust mantiss, Y is display copy
         Copy Sexp to Dexp
         Set SignYexp
         Clr X9
         Break to FINISH
   ClrYPh2          // Result was 0
   Clr SignX
Else                    // Positive exponent
   While (not XDpOvrfl)  // compute Dexp=7-XDp
      XYDexpCnt ++
      XDPCnt ++
   While (not DecEven) // Restore XDp and compute exp in Sexp
      XDpCnt - -              //    It will overshoot by 1
      XYSexpCnt ++
      XYDexpCnt - -
   XYSexp ++               // Fix overshoot
   Copy Sexp to Dexp
   Exch Xreg <> Yreg
FINISH  Clr X9
If SignX then
   Set SignY
Set EOPh1
JMP DSKY
```

BCD Addition

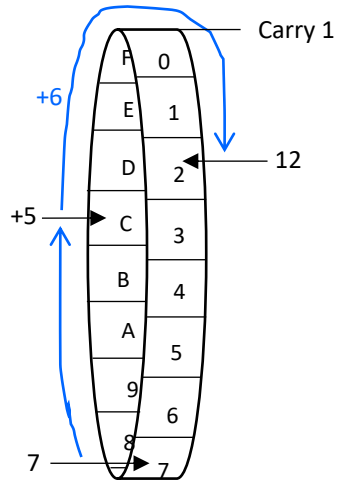
Given a 4-bit adder:

$A_{10} + B_{10} = (A_{BCD} + B_{BCD} + C) \bmod 16 = S_{16}$

$[(A_{BCD} + B_{BCD} + C) \bmod 16 >= 16] = Co$

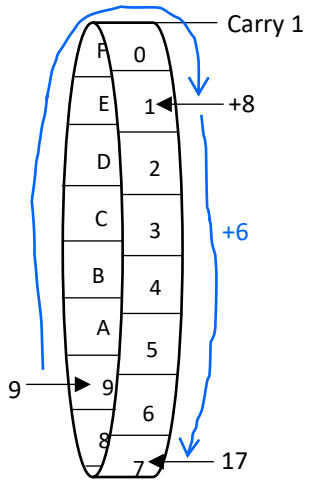
		Co		Ci	Co		Ci	Co		Ci	Co		
7693			0111	1		0110	1		1001			0011	addend
+9684			+1001			+0110			+1000			+0100	addend
17377		1	0001			1101		1	0001			0111	sum1
			+0110			+0110			+0110			+0000	bdc correction
		1	0111		1	0011			0111			0111	sum2

Decimal addition translated to base 16 wheel:



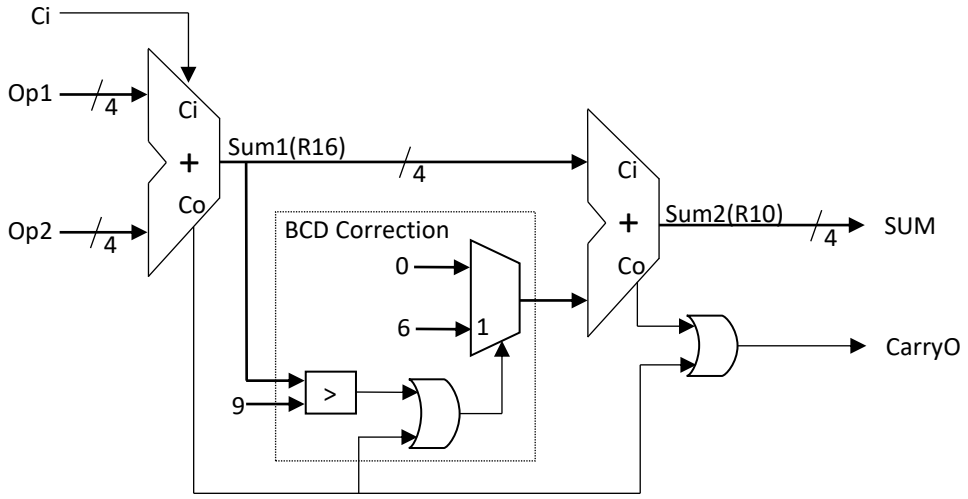

```
  7
+ 5
-----
0xC      If Sum>9 Then
+ 6        Add 6 to Sum
-----
12 bcd
```

Decimal addition translated to base 16 wheel:




```
  9
+ 8
------
0x11     If Hex Rollover
 + 6       Add 6 to Sum
------
17 bcd
```

Op1+Op2+Ci	Sum_{16}	Co	Sum_{10}	C
0	0	0	0	0
1	1	0	1	0
2	2	0	2	0
3	3	0	3	0
4	4	0	4	0
5	5	0	5	0
6	6	0	6	0
7	7	0	7	0
8	8	0	8	0
9	9	0	9	0
10	A	0	0	1
11	B	0	1	1
12	C	0	2	1
13	D	0	3	1
14	E	0	4	1
15	F	0	5	1
16	0	1	6	1
17	1	1	7	1
18	2	1	8	1
19	3	1	9	1



10's Complement

Number Complements: A way of achieving subtraction on a machine capable only of addition

- For decimal arithmetic, 10's complement is the easiest for hardware to do

9's complement of N = N^9:

for each digit N_i in N

$N^9_i = 9 - N_i$

Ex: 9's complement of 5370

```
  9 9 9 9
 -5 -3 -7 -0
  4 6 2 9
```

10's complement of $N = N^{10} = N^9 + 1$:

Ex: 10's complement of 5370

```
  9 9 9 9
 -5 -3 -7 -0
  4 6 2 9
 +       1
 ---------
  4 6 3 0
```

$A - B = A + B^{10}$

How to get the 9's complement of a number with a 4 bit binary adder:

- For each digit N_i in N; Compute the 15's complement and subtract 6 to account for the additional base 16 numerals

$N^9_i = N^{15}_i - 6 = (15 - N_i) - 6$

Conveniently the 15's complement of a BCD digit is also the 1's complement of the BCD digit:

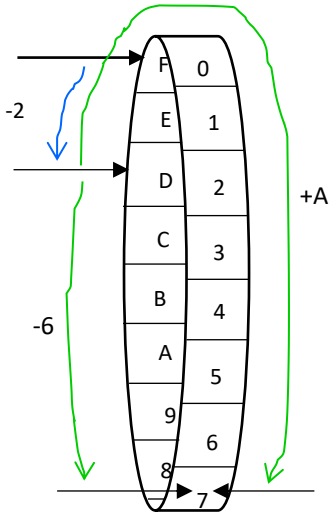
$N^{15}_{BCD} = N^1_{BCD}$; which is just the inverted bits of N_{BCD}

And $(N^{15} - 6) \bmod 16 = (N^{15} + 10) \bmod 16$

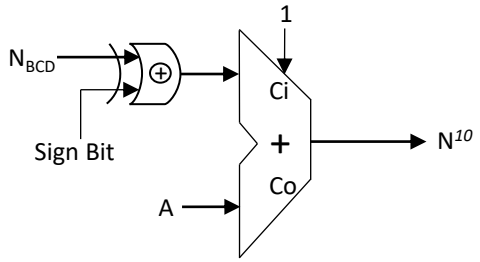
So for each digit in a BCD number N

$N^9_i = \sim N_i + 10$ (~ is bitwise invert)

9s complement of 2 translated to base 16 wheel: (F-2)-6 = (F-2)+A = 7

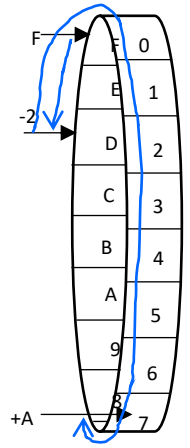


10's Complementer



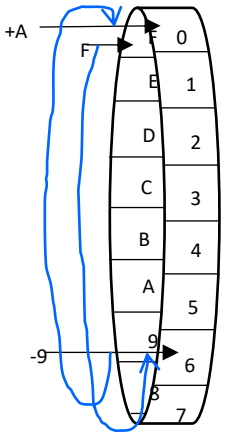
10's complement of 290

9-2 = (F-2+10)mod16 = (~2+A)mod16



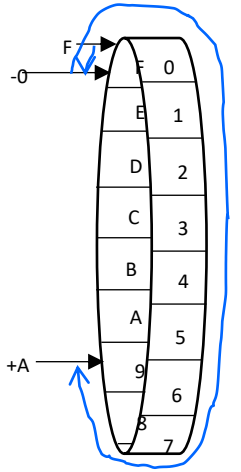
7

F-9+10 = (~9+A)mod16



0

F-0+10 = (~0+A)mod16



9

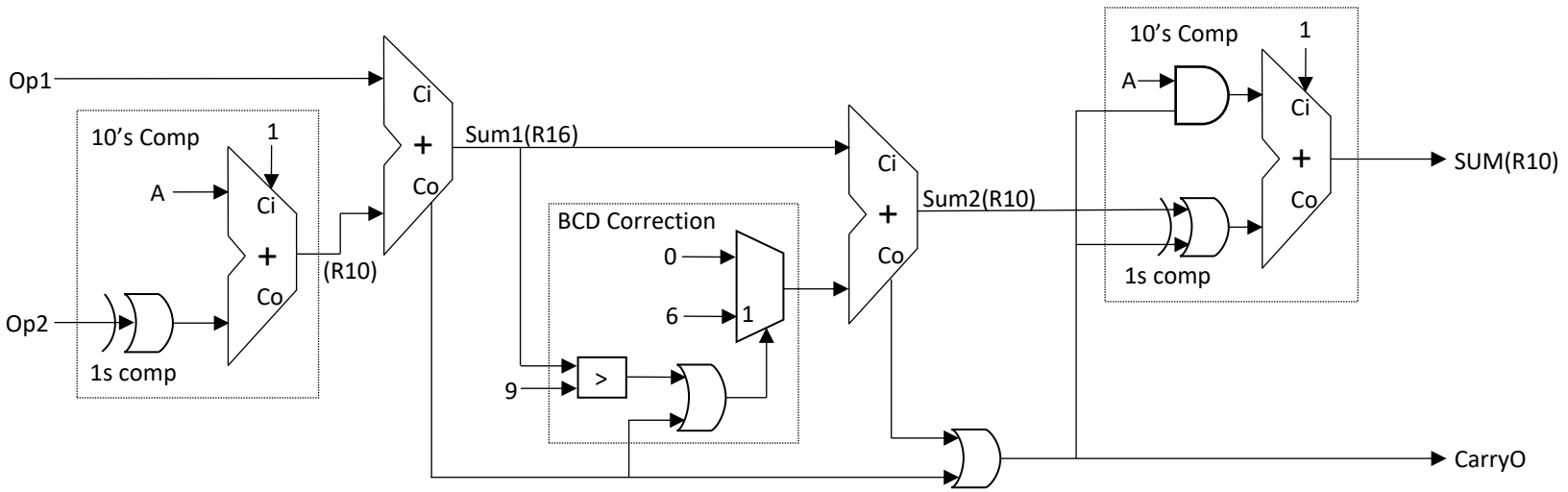
709
+ 1
---
710 = 10's complement of 290

BCD Subtraction

			Co		Ci	Co		Ci	
A	47			0100			0111		minuend
B	-28			-0010			-1000		subtrahend
	19								
				1101			0111		1's complement of subtrahend
				+1010			+1010		add 10
			1	0111		1	0001		9's complement of subtrahend
				+0100			+0111	1	add minuend + 1
				1011			1001		difference
				+0110			+0000		bcd correction
			1	0001			1001		
			Final carry=1 difference is pos (A>=B)						

			Co		Ci	Co		Ci	
A	28			0010			1000		minuend
B	-41			-0100			-0001		subtrahend
	-13								
				1011			1110		1's complement of subtrahend
				+1010			+1010		add 10
			1	0101	1	1	1000		9's complement of subtrahend
				+0010		1	+1000	1	add minuend + 1
			0	1000			0001		difference
				+0000			+0110		bcd correction
				1000			0111		
			Final carry=0 difference is neg (A<B)						
				0111			1000		1's complement
				+1010			+1010		add 10
				0001		1	0010		9's complement
				+0000			+0000	1	add 1
				0001			0011		

Op1-Op2	Co	A + 10#(B)	SUM
-9	0	1	-9
-8	0	2	-8
-7	0	3	-7
-6	0	4	-6
-5	0	5	-5
-4	0	6	-4
-3	0	7	-3
-2	0	8	-2
-1	0	9	-1
0	0	A	0
1	0	B	1
2	0	C	2
3	0	D	3
4	0	E	4
5	0	F	5
6	1	0	6
7	1	1	7
8	1	2	8
9	1	3	9



DexpCnt For Add/Sub Overflow and Insignificant Addend (IA)

		pos/pos								neg/pos							
Xreg(Op2) →		10000000	1000000	100000	10000	1000	100	10	1	0	0	0	0.0001	0.00001	0.000001	0.0000001
Yreg(Op1) ↓		**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**	**-1**	**-2**	**-3**	**-4**	**-5**	**-6**	**-7**
10000000	**7**	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1000000	**6**	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13
100000	**5**	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
10000	**4**	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	11
1000	**3**	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10
100	**2**	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9
10	**1**	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8
1	**0**	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
0.1	**-1**	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
0.01	**-2**	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5
0.001	**-3**	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4
0.0001	**-4**	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3
0.00001	**-5**	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2
0.000001	**-6**	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1
0.0000001	**-7**	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0

pos/neg

neg/neg

IA

IA

Must detect that the insignificant Addend condition boundary crossing during second operand entry and set a bit.
Then at the start of and add, the exponent of the smallest number has to be backed out of DexpCnt and replaced with 0.

Overflow

Overflow happens when Sum generates a carryO of MSD.

Final DexpCnt

-14	0010
-13	0011
-12	0100
-11	0101
-10	0110
-9	0111
-8	1000
-7	1001
-6	1010
-5	1011
-4	1100
-3	1101
-2	1110
-1	1111
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110

IA boundary

DexpCnt Direction

DexpCnt Direction

IA boundary

Multiplication and Division Overflows and Underflows

OP2

A div C...		0.0000001 -7	0.0000001 -7	0.000001 -6	0.000001 -6	0.00001 -5	0.00001 -5	0.0001 -4	0.0001 -4	0 -3	0 -3	0.01 -2	0.01 -2	0.1 -1	0.1 -1	1 0	1 0	10 1	10 1	100 2	100 2	1000 3	1000 3	10000 4	10000 4	100000 5	100000 5	1000000 6	1000000 6	10000000 7	10000000 7
		Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp	Dexp	Sexp
0.0000001	-7	0	-14	-1	-13	-2	-12	-3	-11	-4	-10	-5	-9	-6	-8	-7	-7	-8	-6	-9	-5	-10	-4	-11	-3	-12	-2	-13	-1	-14	0
0.000001	-6	1	-13	0	-12	-1	-11	-2	-10	-3	-9	-4	-8	-5	-7	-6	-6	-7	-5	-8	-4	-9	-3	-10	-2	-11	-1	-12	0	-13	1
0.00001	-5	2	-12	1	-11	0	-10	-1	-9	-2	-8	-3	-7	-4	-6	-5	-5	-6	-4	-7	-3	-8	-2	-9	-1	-10	0	-11	1	-12	2
0.0001	-4	3	-11	2	-10	1	-9	0	-8	-1	-7	-2	-6	-3	-5	-4	-4	-5	-3	-6	-2	-7	-1	-8	0	-9	1	-10	2	-11	3
0.001	-3	4	-10	3	-9	2	-8	1	-7	0	-6	-1	-5	-2	-4	-3	-3	-4	-2	-5	-1	-6	0	-7	1	-8	2	-9	3	-10	4
0.01	-2	5	-9	4	-8	3	-7	2	-6	1	-5	0	-4	-1	-3	-2	-2	-3	-1	-4	0	-5	1	-6	2	-7	3	-8	4	-9	5
0.1	-1	6	-8	5	-7	4	-6	3	-5	2	-4	1	-3	0	-2	-1	-1	-2	0	-3	1	-4	2	-5	3	-6	4	-7	5	-8	6
OP1 1	0	7	-7	6	-6	5	-5	4	-4	3	-3	2	-2	1	-1	0	0	-1	1	-2	2	-3	3	-4	4	-5	5	-6	6	-7	7
10	1	8	-6	7	-5	6	-4	5	-3	4	-2	3	-1	2	0	1	1	0	2	-1	3	-2	4	-3	5	-4	6	-5	7	-6	8
100	2	9	-5	8	-4	7	-3	6	-2	5	-1	4	0	3	1	2	2	1	3	0	4	-1	5	-2	6	-3	7	-4	8	-5	9
1000	3	10	-4	9	-3	8	-2	7	-1	6	0	5	1	4	2	3	3	2	4	1	5	0	6	-1	7	-2	8	-3	9	-4	10
10000	4	11	-3	10	-2	9	-1	8	0	7	1	6	2	5	3	4	4	3	5	2	6	1	7	0	8	-1	9	-2	10	-3	11
100000	5	12	-2	11	-1	10	0	9	1	8	2	7	3	6	4	5	5	4	6	3	7	2	8	1	9	0	10	-1	11	-2	12
1000000	6	13	-1	12	0	11	1	10	2	9	3	8	4	7	5	6	6	5	7	4	8	3	9	2	10	1	11	0	12	-1	13
10000000	7	14	0	13	1	12	2	11	3	10	4	9	5	8	6	7	7	6	8	5	9	4	10	3	11	2	12	1	13	0	14

Dexp and Sexp Counter Values From All Combinations of Op1 and Op2

Division Overflows
Anytime Dexp is positive number greater than 7
Within the positive exp1, negative exp2 quadrant

(DPOvrfl=1)(SignYexp=0)(SignXexp=1)

Mult Overflows
Anytime Sexp is positive number greater than 7
Within the positive exp1, positive exp2 quadrant

(DPOvrfl=1)(SignYexp=SignXexp=0)

Div Underflows

(DPOvrfl=1)(SignYexp=1)(SignXexp=0)

Mult Underflows

(DPOvrfl=1)(SignYexp=SignXexp=1)